When Does Numerical Approximation Matter? Equilibrium Feedback and Bias Correction

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Abstract

Numerical accuracy in structural models is typically assessed through solver tolerances or value-function errors, rather than their consequences for parameter estimation. This paper develops a framework linking numerical approximation of structural fixed points to parameter bias. We show that likelihood normalization bounds parameter displacement without requiring derivative accuracy, and derive a score-projection representation that identifies estimation-relevant errors. For dynamic discrete choice models, we establish mixing-based bounds that remain informative at high discount factors. For dynamic games, equilibrium feedback and estimator-specific projections yield different biases for two-step, nested pseudo-likelihood, and full-solution maximum likelihood, with no general ranking of numerical robustness. Building on these results, we develop feasible parameter-specific diagnostics and bias corrections that relax numerical accuracy requirements during optimization. Applying our diagnostic to Rust's (1987) bus engine data reveals a discretization bias of approximately three percent in the maintenance-cost coefficient, small relative to sampling uncertainty in the original sample. In a dynamic entry game calibrated to Aguirregabiria and Mira (2007), five-point grids generate biases exceeding one standard error at empirically relevant sample sizes, while ten-point grids substantially reduce them.

Keywords: approximated fixed point; nested fixed point; nested pseudo-likelihood; dynamic games; discretization; approximation bias; bias correction.
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1 Introduction

Structural estimation often requires solving a numerical model inside an optimization routine. State spaces are discretized (Rust, 1987), value functions are interpolated or approximated by sieves (Keane and Wolpin, 1994; Arcidiacono et al., 2013; Kristensen et al., 2021), and equilibrium mappings in dynamic games are evaluated numerically (Aguirregabiria and Mira, 2007; Pesendorfer and Schmidt-Dengler, 2008). Even when the optimization converges accurately, the underlying structural model remains approximate. As statistical precision increases, numerical errors that appear small relative to the model solution may become consequential for parameter estimation and inference. Yet conventional accuracy measures, such as Bellman residuals, value-function errors, and solver tolerances, do not directly indicate how much the estimated parameters are affected. The question is therefore not simply how accurately a structural model has been solved, but which numerical errors matter for estimation and how their effects depend on the model and estimator.

This distinction is particularly important in dynamic games. The same approximate equilibrium mapping may be used by two-step, nested pseudo-likelihood (NPL), and full-solution maximum likelihood (ML) estimators, yet the resulting parameter biases need not be similar. Strategic interaction can amplify or attenuate numerical errors, while different estimators respond to different components of those errors. Consequently, an estimator that is less sensitive to approximation in one strategic environment may be more sensitive in another, even when the numerical method is held fixed. Understanding this variation requires tracing numerical errors through the equilibrium mapping and into the estimating equations, rather than examining the accuracy of the computed equilibrium alone.

The consequences of numerical approximation for structural estimation have received important attention. Fernández-Villaverde et al. (2006) (FRS) demonstrate that small errors in computed policy functions can generate substantial likelihood discrepancies that accumulate with sample size. Ackerberg et al. (2009) (AGH) subsequently correct Proposition 2 of FRS and show that, under suitable regularity and approximation conditions, the resulting parameter displacement is of the same order as the approximation error. These findings distinguish discrepancies in the likelihood from displacement of its maximizer. However, general approximation-error bounds do not by themselves identify which numerical errors drive parameter bias, how those errors propagate through structural fixed points, or why alternative estimators may respond differently to the same approximation.

This paper addresses these questions by linking primitive numerical errors to parameter-specific estimation bias in dynamic discrete choice models and dynamic games. The analysis separates three steps: the error introduced by approximating the structural operator, its propagation through the fixed-point solution, and its projection onto the parameter-sensitive directions of the estimation criterion. This distinction is useful because numerical errors

that are large in the underlying model need not generate large parameter biases, while comparatively small errors can materially affect inference when they align with the directions identified by the data.

We begin with a general likelihood result. Under normalization and local identification, uniform accuracy of the approximated log-likelihood alone bounds the pseudo-true parameter displacement at the same rate as the likelihood approximation error, without requiring corresponding accuracy of its derivatives. This refines the sufficient conditions for parameter stability considered by AGH. By contrast, uniform control of a generic extremum criterion generally yields only a square-root bound on parameter displacement. Beyond this rate result, the leading likelihood-based approximation bias is characterized by the projection of the log-likelihood distortion onto the score. This has the familiar interpretation of local misspecification sensitivity (Andrews et al., 2017; Bonhomme and Weidner, 2022). Our focus is on exploiting the structural model to identify the numerical errors that generate this projection, rather than treating the likelihood distortion as an unspecified perturbation.

For single-agent dynamic discrete choice models, we trace numerical errors through the Bellman operator. Conventional contraction arguments bound value-function errors using a factor proportional to $(1-\beta)^{-1}$, where β is the discount factor. Such bounds can be conservative for estimation because choice probabilities depend on differences in choice-specific values, so common components of numerical error cancel. Building on the relative-value perspective of Deaner (2020), we use the multi-step mixing properties of the controlled Markov process to bound the propagation of Bellman residuals into likelihood-relevant errors. These bounds can remain informative as β approaches one. In our engine-replacement experiment, increasing β from 0.9 to 0.999 raises the sup-norm value-function error from approximately 0.29 to 17.4, while the within-state error remains nearly unchanged. The resulting parameter bias also varies little. Thus, numerical accuracy requirements based on conventional value-function bounds may be considerably more stringent than those needed for parameter estimation.

Dynamic games introduce an additional channel through equilibrium feedback. Let $\Psi(P,\theta)$ denote the equilibrium best-response mapping, where P is the CCP profile and θ is the vector of structural parameters. Let Ψ_P denote its Jacobian with respect to P at equilibrium. A primitive numerical residual r_j generates a first-order equilibrium error $(I-\Psi_P)^{-1}r_j$, provided the equilibrium is locally regular. This sensitivity calculation follows from the fixed-point structure, but its implications for estimation differ across methods. We show that two-step estimation projects the one-step residual onto the parameter-sensitive directions of the best-response mapping, whereas NPL and full-solution ML depend on the propagated equilibrium error through different projections. In particular, ML involves an orthogonal projection under the likelihood metric, while NPL generally involves an oblique

projection. Although these estimators build on the established CCP estimation framework (Aguirregabiria and Mira, 2007; Kasahara and Shimotsu, 2012), the resulting comparison characterizes a distinct source of estimator sensitivity: the way a common numerical approximation affects their pseudo-true parameters.

These differences imply that there is no general ranking of two-step, NPL, and ML estimators by robustness to numerical approximation. In the single-agent policy-iteration benchmark, their first-order biases coincide, but strategic interaction breaks this equivalence. Our dynamic entry-game experiment illustrates the resulting variation. Even with moderate equilibrium feedback, the same approximate mapping produces appreciably different parameter biases across estimators. Depending on the strategic environment and parameter, the relative magnitudes change, and some biases have opposite signs. The first-order formulas closely predict these differences, demonstrating that the direction of numerical error and its estimator-specific projection can be as important as its overall magnitude.

The bias representations also yield a practical correction tailored to structural likelihood models. We propose a plug-in adjustment that removes the leading approximation bias using differences between coarse and reference choice probabilities and derivatives of the coarse model. Unlike Newton-type refinements, which require the reference-model score, our correction uses only accurate choice probabilities evaluated at the coarse estimate, avoiding differentiation of the computationally expensive reference solution. The same adjustment also provides a parameter-specific diagnostic of numerical bias relative to sampling uncertainty. Building on the general bias-correction framework of Kristensen and Salanié (2017), we establish second-order accuracy under suitable conditions and compare our procedure with their Newton-type correction. Both methods permit relatively coarse approximations during optimization, relaxing the required accuracy from $\sqrt{T}\delta \to 0$ to $\sqrt{T}\delta^2 \to 0$ when approximation errors in likelihood levels and derivatives have comparable rates. The plug-in correction is particularly attractive when accurate probability evaluations are available but reference-model derivatives are costly.

The same calculations provide a diagnostic for deciding whether numerical approximation is sufficiently accurate for inference. Instead of relying exclusively on value-function or likelihood discrepancies, the researcher can estimate the numerical bias of each parameter and compare it with its standard error. Our Monte Carlo results show why this distinction matters. At a fixed numerical resolution, approximation bias remains essentially unchanged as sample size increases, while confidence interval coverage can deteriorate sharply because sampling uncertainty decreases. The local corrections remove most of the displacement and substantially improve coverage without repeated high-accuracy optimization. These findings suggest treating numerical resolution as a parameter-specific inferential choice rather than solely as a computational tolerance.

Two further exercises demonstrate the practical relevance of our diagnostic. First, revisiting the data and specification of Rust (1987), we replicate his estimates and evaluate a continuous-mileage reference model at the estimated parameters. We find that his 90-state discretization biases the maintenance-cost parameter upward by approximately three percent, consistent with evaluating costs at the lower edge of each mileage bin. This bias amounts to only 0.14 standard errors in the original sample but would reach one standard error at roughly 220,000 bus-months, a scale attainable in modern fleet datasets. Second, in a dynamic entry game based on the payoff structure of Aguirregabiria and Mira (2007), a five-point market-size grid generates biases of 13–17% in the market-size coefficient and 8–10% in the fixed-cost coefficient across the three estimators. These biases exceed one standard error with fewer than 1,000 market-periods, whereas a ten-point grid increases that threshold by roughly two orders of magnitude. These findings illustrate why numerical accuracy should be assessed in parameter units relative to sampling uncertainty.

Our analysis complements several related literatures. General studies of numerical approximation in dynamic models examine the accuracy and stability of computed solutions (Rust, 1996; Santos and Vigo-Aguiar, 1998; Santos, 2000; Stachurski, 2008; Norets, 2010; Kristensen et al., 2021). Deaner (2020) develops relative-value bounds and set-based inference for dynamic discrete choice models, while Kristensen and Salanié (2017) derive higher-order expansions and corrections for approximate estimators. Related corrections in differentiated-products demand estimation are developed by Freyberger (2015). Our analysis differs by connecting primitive fixed-point errors to the parameter displacement induced by a particular approximation and, in dynamic games, by characterizing how this displacement varies across estimators. The score-projection interpretation also connects to the broader literature on sensitivity analysis and locally robust estimation (Kitamura et al., 2013; Armstrong and Kolesár, 2021; Chernozhukov et al., 2022).

Finally, the paper is related to computational methods designed to reduce the cost of solving or estimating dynamic structural models. These include equilibrium computation in dynamic games (Pakes and McGuire, 1994; Doraszelski and Judd, 2012), conditional choice probability methods (Hotz and Miller, 1993), NPL (Aguirregabiria and Mira, 2002, 2007), minimum distance (Pesendorfer and Schmidt-Dengler, 2008), and MPEC (Su and Judd, 2012; Iskhakov et al., 2016). More recently, Adusumilli and Eckardt (2025) develop locally robust estimation based on temporal-difference approximations. Rather than proposing another method for solving or avoiding structural fixed points, we examine how numerical errors in a given approximate model affect alternative estimators and when those errors are consequential for inference.

The remainder of the paper is organized as follows. Section 2 introduces the framework. Section 3 establishes the likelihood approximation bounds and score-projection represen-

tation. Section 4 characterizes error propagation in dynamic discrete choice models and dynamic games. Section 5 develops the asymptotic implications, and Section 6 presents the bias corrections and diagnostics. Section 7 reports the numerical experiments. Proofs are collected in the Appendix.

2 Framework

We develop a general conditional likelihood framework that encompasses the dynamic discrete choice models and dynamic games studied below. The framework allows us to distinguish the statistical effects of numerical approximation from the fixed-point mechanisms that generate them. Let $\{W_t = (Y_t, X_t)\}_{t=1}^T$ be an i.i.d. sample, with realizations $w_t = (y_t, x_t)$. Write $W = (Y, X)$ for a generic random observation and $w = (y, x)$ for its realization. Here, Y_t denotes an outcome, typically a discrete action, and X_t denotes conditioning variables or states with distribution f_X. We treat f_X as fixed when evaluating the conditional likelihood, without requiring the state distribution generated by the structural model to be parameter-independent. The model specifies a conditional density $p(y \mid x; \theta)$ with respect to a σ-finite measure μ (the counting measure for discrete choices), where $\theta \in \Theta \subset \mathbb{R}^k$ and θ_0 denotes the true parameter.[1]

A central feature of these models is that evaluating $p(y \mid x; \theta)$ requires solving for a value function, policy, or equilibrium characterized by a fixed point. Numerical approximation thus affects the likelihood through the underlying fixed-point solution. The following examples illustrate how this structure arises in standard structural estimators and how approximation errors may propagate differently across them.

Example 1 (NFXP for single agent dynamic discrete choice)**.** The nested fixed-point (NFXP) algorithm of Rust (1988) evaluates the likelihood by solving the dynamic programming problem at each candidate parameter value. Consider actions $a \in A = \{0, \ldots, J\}$, states $x \in X$, flow utility $u(x, a; \theta) + \varepsilon(a)$ with i.i.d. type I extreme value shocks (normalized to have mean zero), discount factor $\beta \in (0, 1)$, and controlled transition kernel $f(dx' \mid x, a)$. For each θ, the choice-specific expected continuation value $V_\theta(x, a)$ is the unique fixed point of

$$\Gamma_\theta(V)(x, a) = \int W_\theta(V)(x')\, f(dx' \mid x, a),$$
$$W_\theta(V)(x) = \log \sum_{b \in A} \exp\{u(x, b; \theta) + \beta V(x, b)\}.$$

[1]The framework can accommodate panel data with Markov dependence by using the appropriate conditional likelihood and laws of large numbers and central limit theorems for dependent observations. We adopt the i.i.d. formulation for expositional simplicity.

The resulting conditional choice probabilities (CCPs) are

$$P_\theta(a \mid x) = \frac{\exp\{v_\theta(x,a)\}}{\sum_{b\in A}\exp\{v_\theta(x,b)\}}, \qquad v_\theta(x,a) = u(x,a;\theta) + \beta V_\theta(x,a),$$

as in Rust (1987). Numerical errors in solving the Bellman equation therefore affect estimation through the induced errors in choice-specific values and, ultimately, in the CCPs entering the likelihood.

Example 2 (CCP mappings and dynamic games)**.** Let P denote the vector of conditional choice probabilities (CCPs), either for a single agent or for all players in a dynamic game. The policy iteration or best-response mapping $\Psi(P,\theta)$ yields the optimal CCPs given continuation values computed under P (Aguirregabiria and Mira, 2002, 2007). In a Markov perfect equilibrium, the CCPs satisfy $P = \Psi(P,\theta)$, with the true equilibrium characterized by $P_0 = \Psi(P_0,\theta_0)$. In dynamic games, numerical errors in best responses can propagate through other players' policies, altering the resulting equilibrium CCPs. This strategic feedback gives rise to the estimator-specific effects of numerical approximation examined below.

In dynamic games, let $A = \prod_i A_i$ denote the joint action space. Under conditionally independent private shocks, a CCP profile $P = \{P_i\}_i$ induces the joint probability $P(a \mid x) = \prod_i P_i(a_i \mid x)$, where $a = (a_i)_i$. Thus, $P_j(\theta)$ denotes the approximate profile of player-specific CCPs, while $P_j(a \mid x;\theta)$ denotes its induced joint probability. Throughout, $\sum_{a\in A}$ in likelihood expressions sums over joint actions, whereas equilibrium mappings and their derivatives operate on player-specific CCP profiles. The corresponding Fisher inner product can equivalently be written as a sum over players and their actions, as in Section 4.

Example 3 (Minimum distance)**.** With a finite state space, the minimum distance estimator of Pesendorfer and Schmidt-Dengler (2008) minimizes

$$(\hat{P} - \Psi(\hat{P},\theta))'\hat{\mathcal{W}}(\hat{P} - \Psi(\hat{P},\theta)),$$

where $\hat{P}$ denotes the estimated CCPs and $\hat{\mathcal{W}}$ is a weighting matrix. Although this criterion relies on the same type of numerically approximated fixed-point mapping, it is not likelihood-based. This distinction allows us to identify which properties of approximation error arise from the fixed-point structure and which depend specifically on likelihood normalization.

The preceding examples share a common feature: the likelihood or estimation criterion depends on a fixed-point object that must be approximated numerically. Let j index the numerical approximation, which may involve grid resolution, the number of basis functions or quadrature nodes, interpolation methods, or solver tolerances. More generally, j may represent a combination of numerical procedures.

Replacing the exact operator with $\Gamma_{j,\theta}$ or Ψ_j induces an approximate conditional density $p_j(y \mid x;\theta)$. We assume that $p_j(\cdot \mid x;\theta)$ remains a properly normalized conditional density. This restriction plays a central role in our likelihood-based results: normalization constrains the admissible likelihood distortions, allowing accuracy in the likelihood level alone to yield sharper bounds on parameter displacement than those available for a generic extremum criterion. The population pseudo-true parameter and its sample counterpart are defined by

$$\theta_j = \arg\max_{\theta\in\Theta} Q_j(\theta), \quad Q_j(\theta) = \mathbb{E}[\log p_j(Y \mid X;\theta)],$$
$$\hat{\theta}_j = \arg\max_{\theta\in\Theta} \hat{Q}_j(\theta), \quad \hat{Q}_j(\theta) = \frac{1}{T}\sum_{t=1}^{T} \log p_j(y_t \mid x_t;\theta).$$

Let $Q(\theta) = \mathbb{E}[\log p(Y \mid X;\theta)]$ denote the exact population criterion, uniquely maximized at θ_0.[2] The estimation error can then be decomposed as

$$\hat{\theta}_j - \theta_0 = (\theta_j - \theta_0) + (\hat{\theta}_j - \theta_j),$$

separating the population displacement induced by numerical approximation from the sampling error around the pseudo-true parameter.

We distinguish three measures of numerical approximation error:

$$\delta_j = \sup_{\theta,x,y} \left| \log p_j(y \mid x;\theta) - \log p(y \mid x;\theta) \right|,$$
$$\delta_j^{(k)} = \sup_{\theta\in\mathcal{N},x,y} \left\| \nabla_\theta^k \log p_j(y \mid x;\theta) - \nabla_\theta^k \log p(y \mid x;\theta) \right\|, \qquad k = 1, 2,$$

where $\mathcal{N}$ is a neighborhood of θ_0, and $\|\cdot\|$ denotes the Euclidean norm for vectors and the operator norm for matrices. For convenience, write $\delta_j' = \delta_j^{(1)}$ and $\delta_j'' = \delta_j^{(2)}$. These quantities measure approximation errors in the log-likelihood level, gradient, and Hessian, respectively. While Ackerberg et al. (2009) control $\max\{\delta_j, \delta_j', \delta_j''\}$, we distinguish them because they play different roles in estimation and inference. In particular, we show that likelihood normalization yields an $O(\delta_j)$ bound on parameter displacement without requiring derivative accuracy, whereas sharper bias expansions and asymptotic results impose additional conditions on δ_j' and δ_j''.

Throughout, let $s(w) = \nabla_\theta \log p(y \mid x;\theta_0)$ denote the score of the exact model and $I = \mathbb{E}[s(W)s(W)']$ the Fisher information matrix. Define the log-likelihood approximation error as

$$g_j(y, x;\theta) = \log p_j(y \mid x;\theta) - \log p(y \mid x;\theta),$$

[2]The pseudo-true parameter has the usual interpretation under model misspecification (White, 1982), with the misspecification here induced by numerical approximation.

and write $g_j(w) = g_j(y, x; \theta_0)$ for its evaluation at the true parameter. While $g_j(w)$ captures the magnitude and direction of the likelihood distortion, its projection onto the score determines the leading displacement of the parameter estimate. This distinction forms the basis of the first-order bias characterization developed in the next section.

3 How Large Is the Approximation Bias?

Numerical approximation can affect both the value of the likelihood and the location of its maximizer, but these effects need not be closely related. FRS showed that small errors in computed policy functions can generate substantial likelihood discrepancies that accumulate with sample size, raising concerns about the reliability of numerical approximations in structural estimation. In contrast, Ackerberg et al. (2009) corrected FRS's Proposition 2 and established that, under suitable regularity conditions, the displacement of the maximizer is of the same order as the underlying approximation error.

These findings are best understood as addressing distinct aspects of numerical approximation. While FRS highlight discrepancies between the exact and approximated statistical models, AGH establish that such discrepancies need not translate into comparable errors in parameter estimates. Their result, however, characterizes only the order of parameter displacement, leaving open which components of numerical error matter for estimation and how the underlying fixed-point structure shapes their effects. This section addresses these questions in two steps. We first show that likelihood normalization yields an $O(\delta_j)$ bound on parameter displacement using only the accuracy of the likelihood level, a result that does not generally extend to other extremum criteria. We then characterize the direction and magnitude of the leading approximation bias through its projection onto the score.

3.1 Level control suffices for likelihood-based estimators

We begin by examining how uniform accuracy of the approximated likelihood translates into parameter accuracy. Unlike a generic extremum criterion, a likelihood is subject to the normalization constraint that its underlying density integrates to one. We show that this additional structure yields an $O(\delta_j)$ bound on parameter displacement without requiring accuracy of the likelihood derivatives. We first state the regularity conditions.

Assumption 1. (i) The parameter space $\Theta \subset \mathbb{R}^k$ is compact, with $\theta_0 \in \operatorname{int}\Theta$. (ii) The population criterion Q is continuous and uniquely maximized at θ_0. Thus, for every $\varepsilon > 0$, there exists $\eta(\varepsilon) > 0$ such that $\sup_{\theta\in\Theta:\,\|\theta-\theta_0\|\geq\varepsilon} Q(\theta) \leq Q(\theta_0) - \eta(\varepsilon)$ whenever the set is nonempty. (iii) There exist $\varepsilon_0, \lambda > 0$ such that $Q(\theta_0) - Q(\theta) \geq \lambda\|\theta - \theta_0\|^2$ for all $\theta \in \Theta$ satisfying $\|\theta - \theta_0\| \leq \varepsilon_0$. (iv) The conditional density is locally L^1-Lipschitz in θ: for

some $L < \infty$, $\mathbb{E}_X\big[\int |p(y \mid X;\theta) - p(y \mid X;\theta_0)|\, d\mu(y)\big] \le L\|\theta - \theta_0\|$ whenever $\theta \in \Theta$ and $\|\theta - \theta_0\| \le \varepsilon_0$. (v) For every j, $\theta \in \Theta$, and x, $p_j(\cdot \mid x;\theta)$ is a properly normalized probability density with respect to μ, and $\delta_j \le 1$.

Part (iii) imposes standard local quadratic identification, which holds, for example, when Q is twice continuously differentiable near θ_0 and $-\nabla^2 Q(\theta_0) = I$ is positive definite. Part (iv) requires local L^1 continuity at a linear rate, as implied by standard differentiability in quadratic mean. These conditions concern the exact statistical model rather than its numerical approximation. The key restriction on the approximation, imposed in part (v), is that it preserves the normalization of the conditional density. As the following theorem establishes, this structure allows uniform accuracy of the log-likelihood level alone to yield an $O(\delta_j)$ bound on parameter displacement, without requiring accuracy of its derivatives.

Theorem 1. *Under Assumption 1, if* $2\delta_j < \eta(\varepsilon_0)$*, then*

$$\|\theta_j - \theta_0\| \le K\delta_j, \qquad K = \frac{L + \sqrt{L^2 + 2e\lambda}}{2\lambda}.$$

The key to the result is likelihood normalization. Define $g_j(y, x;\theta) = \log p_j(y \mid x;\theta) - \log p(y \mid x;\theta)$, so that $|g_j(y, x;\theta)| \le \delta_j$. Since $p_j = p\, e^{g_j}$ and both densities integrate to one, $\int p\, g_j\, d\mu = -\int p(e^{g_j} - 1 - g_j)\, d\mu = O(\delta_j^2)$, where the densities and g_j are evaluated at a common (x, θ). Thus, under the exact model at any θ, the average log-likelihood distortion is second order in the approximation error. It follows that the population criterion discrepancy can be expressed as

$$Q_j(\theta) - Q(\theta) = \mathbb{E}_X \int \{p(y \mid X;\theta_0) - p(y \mid X;\theta)\} g_\theta\, d\mu + O(\delta_j^2) = O(\delta_j\|\theta - \theta_0\| + \delta_j^2).$$

The first term vanishes as θ approaches θ_0, while the remaining term is second order in δ_j. Combined with local quadratic identification, this structure yields the $O(\delta_j)$ bound on parameter displacement.

Theorem 1 provides a likelihood-specific refinement of the parameter stability result of Ackerberg et al. (2009). Under normalization and local identification, uniform control of the log-likelihood approximation error alone yields a linear bound on parameter displacement, without requiring corresponding control of the approximation error in its derivatives. This distinction is useful when numerical function values are accurate but derivative approximations are difficult to control.

The result also helps reconcile the different messages in FRS and Ackerberg et al. (2009). The likelihood evaluated on a sample can differ substantially from its exact counterpart even when the location of its maximizer changes little. Normalization restricts precisely the component of the population likelihood distortion that can move the maximizer to first order.

Thus, the accumulation of likelihood discrepancies emphasized by FRS remains relevant for assessing the numerical representation of the statistical model, likelihood comparisons, and some likelihood based diagnostics, but it is not by itself a measure of approximation bias in the parameter estimate. For that purpose, the local shape and direction of the distortion matter. The next proposition also shows that the favorable $O(\delta_j)$ conclusion is specific to likelihood structure and does not follow from level accuracy of an arbitrary criterion.

Proposition 1. *Let $\bar{Q}$ satisfy Assumption 1(ii)–(iii), with λ, ε_0, and η defined for $\bar{Q}$. Suppose $\bar{Q}_j$ satisfies $\sup_{\theta\in\Theta} |\bar{Q}_j(\theta) - \bar{Q}(\theta)| \leq \delta$. Then any maximizer $\bar{\theta}_j$ of $\bar{Q}_j$ satisfies*

$$\|\bar{\theta}_j - \theta_0\| \leq (2\delta/\lambda)^{1/2}$$

whenever $2\delta < \eta(\varepsilon_0)$. This rate is sharp in general.

The square-root bound is a standard consequence of uniform objective approximation and local quadratic identification. We include it to highlight the additional protection provided by likelihood normalization; see also Dubé et al. (2012) for related concerns about numerical accuracy in GMM estimation. To establish sharpness, consider $\Theta = [-1, 1]$, $\bar{Q}(\theta) = -\theta^2/2$, and

$$\bar{Q}_j(\theta) = \bar{Q}(\theta) + \delta\phi\left((\theta - \sqrt{\delta})/\sqrt{\delta}\right), \qquad \phi(u) = \max\{0, 1 - 2|u|\}. \tag{1}$$

For sufficiently small $\delta > 0$, $\sup_{\theta\in\Theta} |\bar{Q}_j(\theta) - \bar{Q}(\theta)| = \delta$, while the unique maximizer is $\bar{\theta}_j = \sqrt{\delta}$. Thus, a perturbation uniformly bounded by δ can shift the maximizer by exactly $\sqrt{\delta}$, establishing sharpness. The perturbation is concentrated in a neighborhood of width $O(\sqrt{\delta})$, with slopes of order $O(\sqrt{\delta})$. Unlike a generic objective, likelihood normalization restricts the admissible distortions and yields the sharper $O(\delta_j)$ bound in Theorem 1.

For minimum distance and GMM estimators, an $O(\delta_{m,j})$ parameter bound can instead be obtained by controlling approximation error at the level of the moment function. Let $m(\theta)$ denote the population moment function, with $m(\theta_0) = 0$, and let $m_j(\theta)$ be its numerical approximation satisfying $\sup_{\theta\in\Theta} \|m_j(\theta) - m(\theta)\| \leq \delta_{m,j}$. If $\bar{\theta}_j$ minimizes $\|m_j(\theta)\|^2$, then $\|m_j(\bar{\theta}_j)\| \leq \|m_j(\theta_0)\| \leq \delta_{m,j}$, implying $\|m(\bar{\theta}_j)\| \leq 2\delta_{m,j}$. Under a suitable local identification condition, this yields $\|\bar{\theta}_j - \theta_0\| = O(\delta_{m,j})$.

By contrast, controlling only the level of a generic objective yields at best an $O(\delta^{1/2})$ parameter bound, requiring $\sqrt{T}\delta^{1/2} \to 0$, or equivalently $T\delta \to 0$, for numerical approximation to be asymptotically negligible at the $\sqrt{T}$ scale. Thus, a sample-size related concern analogous to that emphasized by FRS can arise for non-likelihood criteria, although for a different reason: the weaker information provided by objective-level accuracy, rather than the accumulation of discrepancies in the joint likelihood.

3.2 The first-order bias

The preceding results establish the order of the pseudo-true parameter displacement but do not characterize its direction. Two numerical approximations of comparable magnitude can have markedly different effects on parameter estimates, depending on whether their errors align with or are orthogonal to parameter-sensitive directions of the model. The following result characterizes this relationship through a score projection, linking numerical approximation to local misspecification.

To characterize the leading approximation bias, we impose additional smoothness and derivative accuracy conditions on the exact and approximated likelihoods.

Assumption 2. (i) The functions $\theta \mapsto \log p(y \mid x;\theta)$ and $\theta \mapsto \log p_j(y \mid x;\theta)$ are twice continuously differentiable on $\mathcal{N}$, with their derivatives dominated by an integrable envelope. Moreover, $\nabla^2_\theta Q$ is Lipschitz continuous on $\mathcal{N}$, and $I = -\nabla^2_\theta Q(\theta_0)$ is positive definite. (ii) $\mathbb{E}[\|s(W)\|] < \infty$. (iii) As $j \to \infty$, $\delta_j \to 0$, $\delta'_j \to 0$, and $\delta''_j \to 0$.

Theorem 2. *Under Assumptions 1–2,*

$$\theta_j - \theta_0 = B_j + O\big(\delta_j(\delta_j + \delta'_j + \delta''_j)\big),$$
$$B_j = -I^{-1}\mathbb{E}[s(W)g_j(W)] = -I^{-1}\mathbb{E}\left[s(W)\frac{\pi_j(Y \mid X)}{p(Y \mid X;\theta_0)}\right] + O(\delta_j^2),$$

where $\pi_j(y \mid x) = p_j(y \mid x;\theta_0) - p(y \mid x;\theta_0)$. *In particular,* $\|B_j\| \le \delta_j \|I^{-1}\|\mathbb{E}[\|s(W)\|]$.

Theorem 2 characterizes not only the order of approximation bias but also its direction. Since $\mathbb{E}[s(W)] = 0$, the leading bias $B_j = -I^{-1}\mathbb{E}[s(W)g_j(W)]$ is the negative of the population linear projection coefficient of the log-likelihood distortion $g_j(W)$ on the score $s(W)$. To make this explicit, consider the projection

$$g_j(W) = s(W)'c_j + \nu_j(W), \qquad c_j = I^{-1}\mathbb{E}[s(W)g_j(W)],$$

where $\mathbb{E}[s(W)\nu_j(W)] = 0$ by construction. The component $s(W)'c_j$ lies in the tangent space spanned by the score and can be offset locally by a parameter adjustment of $-c_j$. Accordingly, the pseudo-true parameter shifts by $\theta_j - \theta_0 = -c_j$ to first order, while the orthogonal component $\nu_j(W)$ has no first-order effect on parameter bias.

The projection has the familiar interpretation of a local misspecification effect (Andrews et al., 2017; Bonhomme and Weidner, 2022). The contribution here is to apply this representation to numerical approximation and, in the following section, express the relevant likelihood distortion in terms of primitive fixed-point errors. This connection allows the direction of approximation bias to be characterized using numerical objects available from the structural model.

This decomposition also clarifies the distinction between likelihood discrepancies and parameter displacement. Numerical approximation may substantially distort the likelihood while having little effect on parameter estimates if the distortion is largely orthogonal to the score. Conversely, even a small distortion can generate meaningful bias when it aligns with parameter-sensitive directions of the model. The relevant measure of approximation error for point estimation is therefore its projection onto the score, rather than its overall magnitude. Components orthogonal to the score may nevertheless affect the maximized likelihood, model comparison, and specification assessment.

For discrete choice models, the score projection can be expressed directly in terms of conditional choice probabilities (CCPs):

$$B_j = -I^{-1}\mathbb{E}_X\left[\sum_{a\in A}\frac{\nabla_\theta P(a \mid X;\theta_0)\,\pi_j(a \mid X)}{P(a \mid X;\theta_0)}\right],\; I = \mathbb{E}_X\left[\sum_{a\in A}\frac{\nabla_\theta P(a \mid X;\theta_0)\nabla_\theta P(a \mid X;\theta_0)'}{P(a \mid X;\theta_0)}\right]. \tag{2}$$

The leading bias is thus determined by a weighted projection of the CCP approximation error π_j onto the CCP derivatives with respect to the structural parameters, using inverse-probability weights $1/P$ corresponding to the Pearson chi-square metric. Only those components of the approximation error that align with parameter-sensitive changes in CCPs contribute to the first-order bias. This representation allows us to assess the relevant errors state by state. Section 4 further traces these errors to primitive numerical residuals and their propagation through dynamic and strategic feedback.

Theorem 2 characterizes the leading approximation bias in terms of the exact-model score and information matrix, which may be computationally costly to evaluate. The following corollary provides a feasible alternative using derivatives of the approximate model and the density discrepancy evaluated at a single parameter value.

Corollary 1 (Approximate-score bias representation)**.** *Let $s_j(w) = \nabla_\theta \log p_j(y \mid x;\theta_0)$ and $I_j = -\nabla^2_\theta Q_j(\theta_0)$ denote the score and negative expected Hessian of the approximate model, respectively. Suppose further that $\nabla^2_\theta Q_j$ is Lipschitz continuous on $\mathcal{N}$ uniformly in j. Then*

$$\theta_j - \theta_0 = \tilde{B}_j + O(\delta_j^2), \qquad \tilde{B}_j = -I_j^{-1}\mathbb{E}_X\left[\int s_j(y,X)\,\pi_j(y \mid X)\,d\mu(y)\right],$$

where $\pi_j(y \mid x) = p_j(y \mid x;\theta_0) - p(y \mid x;\theta_0)$.

Corollary 1 provides a practical bias diagnostic without requiring repeated evaluation of the accurate model during optimization. The correction $\tilde{B}_j$ uses derivatives of the approximate likelihood, which are already available from estimation, together with the density discrepancy π_j evaluated at a single parameter value. The latter requires only one accurate model solution. Importantly, the representation exploits the exact identity

$\nabla_\theta Q_j(\theta_0) = -\mathbb{E}_X[\int s_j(y, X)\pi_j(y \mid X)\, d\mu(y)]$, avoiding the $O(\delta_j\delta_j')$ error associated with replacing the approximate-model score by the exact score in Theorem 2.

This refinement is particularly useful when the leading bias is small because the numerical distortion is nearly orthogonal to the score. In such cases, errors in approximating the score may be substantial relative to the leading projection, reducing the accuracy of the original bias formula. The feasible representation instead retains second-order accuracy. The dynamic game experiments in Section 7 illustrate the quantitative importance of this distinction.

The following location model, adapted from Ackerberg et al. (2009), provides a simple illustration of how substantial likelihood discrepancies can coexist with a small and precisely characterized parameter displacement.

Example 4 (AGH location model)**.** Consider $y = \theta + v$, where $v \sim N(0, \sigma^2)$, and suppose the approximated model is $y = \theta + \Delta + v$ for a fixed numerical error Δ. At the true parameter θ_0, the log-likelihood distortion and exact score are $g_j(w) = \Delta(y - \theta_0)/\sigma^2 - \Delta^2/(2\sigma^2)$ and $s(w) = (y - \theta_0)/\sigma^2$, respectively. The score projection yields $B_j = -I^{-1}\mathbb{E}[s(W)g_j(W)] = -\Delta$, which coincides with the exact parameter displacement $\hat{\theta}_j - \hat{\theta} = -\Delta$, where $\hat{\theta} = \bar{y}$ and $\hat{\theta}_j = \bar{y} - \Delta$. Thus, although discrepancies in the joint likelihood can grow substantially with sample size, as illustrated by Ackerberg et al. (2009), the approximation bias remains fixed at $-\Delta$, independently of T.[3]

Theorem 2 goes beyond the distinction illustrated by the location example by identifying the component of likelihood distortion that determines parameter bias. Specifically, the leading displacement depends on the projection of the distortion onto the score, rather than its overall magnitude. This characterization explains why numerical approximations of comparable accuracy can have markedly different effects on parameter estimates.

The theorem also clarifies the role of derivative accuracy. The leading bias depends on the log-likelihood distortion at θ_0, whereas approximation errors in the score and Hessian contribute only through higher-order terms involving $\delta_j\delta_j'$ and $\delta_j\delta_j''$. Consequently, the likelihood and its derivatives need not be approximated at the same rate for the first-order bias expansion to remain informative. This distinction is particularly relevant for adaptive grids, simulation-based methods, and iterative solvers, where function values may be computed more accurately than their derivatives. Although derivative accuracy remains important for estimating curvature, computing standard errors, and obtaining higher-order refinements, it does not directly enter the leading bias term. The appropriate numerical accuracy requirements therefore depend on the inferential object of interest.

[3]Although the Gaussian example does not satisfy the uniform log-likelihood approximation bound in Assumption 1 due to its unbounded support, the score-projection formula holds exactly in this case, illustrating the underlying bias mechanism beyond the theorem's sufficient conditions.

4 Fixed-Point Structure and Approximation Bias

Theorems 1–2 characterize approximation bias in terms of the induced log-likelihood distortion g_j. In structural models, however, researchers typically control more primitive sources of numerical error, such as discretization, quadrature, interpolation, and solver residuals, rather than g_j directly. These errors arise from approximating the underlying fixed-point problem and can often be evaluated or bounded using the numerical procedure itself.

This section exploits the fixed-point structure to trace how these primitive errors propagate into the likelihood and, ultimately, into parameter bias. We establish two main results. First, in single-agent dynamic discrete choice models, approximation bias need not increase sharply as the discount factor approaches one, because choice probabilities depend on differences in continuation values rather than their levels. Second, in dynamic games, numerical errors propagate through equilibrium feedback, with effects that differ across two-step, nested pseudo-likelihood (NPL), and full-solution maximum likelihood estimators. Thus, the fixed-point structure determines not only the magnitude of the induced likelihood distortion but also its direction and, consequently, its effect on parameter estimates.

4.1 Nested fixed point: discounting and differences

We begin with the nested fixed-point problem in Example 1. Consider an approximate Bellman operator of the form

$$\Gamma_{j,\theta}(V)(x,a) = \int W_\theta(V)(x')\, K_j(dx' \mid x,a), \tag{3}$$

where $K_j(\cdot \mid x,a)$ is a probability measure for each (x,a). This formulation encompasses Rust-style state-space discretization, quadrature with positive weights, and interpolation schemes with nonnegative weights that reproduce constants, including piecewise-constant and piecewise-linear interpolation combined with the transition kernel. In each case, the approximate operator can be represented using a Markov kernel.[4] Evaluating (3) at arbitrary states, rather than only at grid points, provides a natural off-grid extension consistent with likelihood evaluation at observed states.

Let V_j and $V = V_{\theta_0}$ denote the fixed points of Γ_{j,θ_0} and Γ_{θ_0}, respectively. Define the

[4]Higher-order polynomial and spline interpolation, as well as Chebyshev approximation, may involve negative weights and need not preserve the sup-norm contraction property; see Stachurski (2008). Our general bias representation in Theorem 2 does not require this structure, but the discount-factor robust bounds derived below do.

value-function error and the one-step approximation residual as

$$e_j = V_j - V, \qquad r_j = \Gamma_{j,\theta_0}(V) - \Gamma_{\theta_0}(V) = \int W_{\theta_0}(V)(x')\,\{K_j - f\}(dx' \mid x, a).$$

While r_j measures the direct error introduced by the approximate operator, e_j also reflects its propagation through the Bellman fixed point.

To distinguish errors in value-function levels from those relevant for choice probabilities, define the span of a function h on $X \times A$ as $\mathrm{sp}(h) = \sup_{x,a} h(x,a) - \inf_{x,a} h(x,a)$ and its within-state oscillation as

$$\mathrm{osc}(h) = \sup_x \max_{a,a' \in A} |h(x,a) - h(x,a')| \le \mathrm{sp}(h).$$

For a Markov kernel M on $X \times A$, define Dobrushin's ergodicity coefficient as $\tau(M) = \sup_{z,z'} \sup_B |M(B \mid z) - M(B \mid z')| \in [0,1]$, where $z, z' \in X \times A$ and B is a measurable subset of $X \times A$. For finite kernels, this equals half the maximum pairwise row L^1 distance and satisfies $\mathrm{sp}(Mh) \le \tau(M)\,\mathrm{sp}(h)$.[5] The coefficient measures how strongly the Markov kernel contracts differences across initial states.

The following proposition relates the primitive residual r_j to the resulting value-function error and shows that the component relevant for estimation can remain bounded as the discount factor approaches one.

Proposition 2 (NFXP approximation errors)**.** *Consider Example 1 under the approximation in (3), with bounded value functions and residuals.*

(a) Define $\bar{P}_j(\cdot \mid x') = \int_0^1 P(\cdot \mid x'; V + te_j)\,dt$*, where* $P(\cdot \mid x'; U)$ *denotes the logit CCPs induced by the value function* U *at* θ_0*. Let* $\bar{M}_j$ *be the Markov kernel on* $X \times A$ *given by* $\bar{M}_j(dx', a' \mid x, a) = K_j(dx' \mid x, a)\bar{P}_j(a' \mid x')$*. Then*

$$(I - \beta \bar{M}_j)e_j = r_j, \qquad e_j = (I - \beta \bar{M}_j)^{-1} r_j = \sum_{k \ge 0} \beta^k \bar{M}_j^k r_j.$$

Let M_j *denote the corresponding kernel using the exact CCPs* P_{θ_0} *in place of* $\bar{P}_j$*. Then* $e_j = (I - \beta M_j)^{-1} r_j + R_j$*, where* $\mathrm{osc}(R_j) \le C\beta\,\mathrm{osc}(e_j)^2 \sum_{k \ge 0} \beta^k \tau(M_j^k)$*. In particular,* $\mathrm{osc}(R_j) = O(\mathrm{osc}(e_j)^2)$ *for fixed* $\beta < 1$*, uniformly in* β *if the Dobrushin sum for* M_j *is uniformly bounded.*

(b) The induced log-likelihood distortion satisfies $|g_j(a, x; \theta_0)| \le \beta\,\mathrm{osc}(e_j)$ *for all* (x, a)*.*

(c) The value-function error satisfies $\mathrm{osc}(e_j) \le \mathrm{sp}(e_j) \le \mathrm{sp}(r_j) \sum_{k \ge 0} \beta^k \tau(\bar{M}_j^k)$*. If* $\bar{M}_j$ *satisfies an* m*-step Doeblin condition,* $\bar{M}_j^m(\cdot \mid z) \ge \rho\nu(\cdot)$ *for all* $z \in X \times A$ *and some*

[5] See Seneta (1979) for the theory of coefficients of ergodicity.

probability measure ν, then

$$\operatorname{osc}(e_j) \leq \frac{m}{1-\beta^m(1-\rho)} \operatorname{sp}(r_j) \leq \frac{m}{\rho} \operatorname{sp}(r_j).$$

The multiplicative constant is uniform in $\beta \in (0,1)$ if m and $\rho > 0$ can be chosen independently of β; a uniform bound on $\operatorname{osc}(e_j)$ additionally requires $\operatorname{sp}(r_j)$ to remain uniformly bounded.

(d) For any constant c, $\operatorname{sp}(r_j) \leq 2 \sup_{x,a} \left| \int \{W_{\theta_0}(V)(x') - c\} \{K_j - f\}(dx' \mid x, a) \right|$.

The conventional contraction argument yields $\sup |e_j| \leq \sup |r_j|/(1-\beta)$, suggesting that numerical errors may become increasingly severe as $\beta \to 1$. Proposition 2 explains why this bound can be overly conservative for estimation. Part (b) shows that choice probabilities depend only on within-state differences in value-function errors, while part (d) establishes that constant components of continuation values cancel from the primitive residual because both transition kernels preserve probabilities. Thus, the residual depends on errors in integrating relative continuation values rather than their potentially large discounted levels. Part (c) further shows that these errors propagate according to the mixing properties of the controlled Markov process. Under suitable uniform bounds on residual variation and mixing, the likelihood-relevant approximation error need not exhibit the $(1-\beta)^{-1}$ deterioration of conventional sup-norm bounds.

The engine replacement model illustrates this mechanism. Its replacement-reset structure implies a two-step Doeblin condition for the kernel $\bar{M}_j$ in Proposition 2(a). After one transition, the state lies on the grid G, where replacement occurs with probability at least $\rho_j = \min_{x \in G} \bar{P}_j(1 \mid x) > 0$. Conditional on replacement, the subsequent state distribution is independent of the preceding mileage history. Thus, $\bar{M}_j$ satisfies the minorization condition with $m = 2$ and $\rho = \rho_j$. The resulting bound depends on the mixing properties of the controlled process rather than the level of discounted continuation values. The bounds in Proposition 2(c) can differ considerably in tightness. The Dobrushin sum $\bar{C}_j = \sum_{k \geq 0} \beta^k \tau(\bar{M}_j^k)$ captures the decay of dependence on initial states. Although $\bar{M}_j$ operates on a continuous state space, its finite-grid structure provides computable bounds on $\bar{C}_j$.

Remark 1 (Computing the Dobrushin sum). Let T_j denote the restriction of $\bar{M}_j$ to $G \times A$, a stochastic matrix of dimension $|G||A|$. Since $\bar{M}_j(\cdot \mid z)$ is supported on $G \times A$ for every $z \in X \times A$, each row of $\bar{M}_j^k$ with $k \geq 1$ is a convex combination of the rows of T_j^{k-1}:

$$\bar{M}_j^k(\cdot \mid z) = \sum_{g \in G \times A} \bar{M}_j(g \mid z)\, T_j^{k-1}(\cdot \mid g).$$

We use the following elementary fact: if $\{\rho_g\}_g$ is a finite family of probability vectors and

w, w' are probability weights, then $\|\sum_g w_g\rho_g - \sum_g w'_g\rho_g\|_1 \leq \max_{g,g'} \|\rho_g - \rho_{g'}\|_1$. To see this, write $\sum_g w_g\rho_g - \sum_g w'_g\rho_g = \sum_{g,g'} w_g w'_{g'}(\rho_g - \rho_{g'})$, since $\sum_{g'} w'_{g'} = \sum_g w_g = 1$, and apply the triangle inequality. Taking $\rho_g = T_j^{k-1}(\cdot \mid g)$ and $w = \bar{M}_j(\cdot \mid z)$, $w' = \bar{M}_j(\cdot \mid z')$ for arbitrary $z, z' \in X \times A$ gives $\tau(\bar{M}_j^k) \leq \tau(T_j^{k-1})$ for all $k \geq 1$, uniformly over the continuous state space; for $k = 1$ the right-hand side is $\tau(I) = 1$, which is the trivial bound. Conversely, the rows of $\bar{M}_j^k$ indexed by $z \in G \times A$ are exactly the rows of T_j^k, so $\tau(\bar{M}_j^k) \geq \tau(T_j^k)$. Defining $C_j^G = \sum_{k\geq 0} \beta^k \tau(T_j^k)$, we obtain $C_j^G \leq \bar{C}_j \leq 1 + \beta\, C_j^G$. Both bounds are computable from the finite matrix T_j. Moreover, if $\tau(T_j^K) < 1$ for some K, submultiplicativity of τ gives $\tau(T_j^m) \leq \tau(T_j^K)^{\lfloor m/K \rfloor}$, providing a geometric bound on the remaining series tail.

In our experiment, the computable upper bound $1 + \beta C_j^G$ ranges from approximately 3.8 to 4.5 and remains relatively stable as β approaches one. By contrast, the simpler Doeblin bound $m/[1 - \beta^m(1 - \rho_j)]$ is substantially more conservative: with $\rho_j \approx 0.005$, its implied error bound can exceed the realized error by more than two hundred times at high discount factors. Thus, the within-state oscillation $\mathrm{osc}(e_j)$, when available, and the finite-grid Dobrushin bounds provide more informative measures of numerical accuracy than the minorization bound alone (Table 3).

This analysis is closely related to Deaner (2020), who derives bounds on choice-specific value differences using Bellman residuals and total-variation distances, which can also avoid deterioration as $\beta \to 1$. Proposition 2 characterizes residual propagation through multi-step mixing, while our additional step connects the resulting relative-value errors to parameter bias. Combining Proposition 2 with Theorem 2, let $h_j = (I - \beta M_j)^{-1} r_j$. The leading approximation bias satisfies

$$B_j = -\beta I^{-1} \mathbb{E}_X \left[\sum_{a \in A} \nabla_\theta P(a \mid X; \theta_0) h_j(X, a) \right] + O(\mathrm{osc}(e_j)^2).$$

The resolvent propagates the primitive residual through the controlled Markov process, within-state differences determine the resulting likelihood distortion, and the score projection isolates the component relevant for parameter bias.

4.2 CCP mappings and dynamic games

We next turn to CCP mappings in dynamic games, where numerical errors propagate through strategic best responses and equilibrium feedback (Pakes and McGuire, 1994; Aguirregabiria and Mira, 2007; Pesendorfer and Schmidt-Dengler, 2008). For simplicity, consider a finite state space X, with extensions to continuous states understood through the corresponding integral operators. Let $P = \{P_i(a_i \mid x)\}_{i,a_i,x}$ denote the profile of players' CCPs, where $\sum_{a_i} P_i(a_i \mid x) = 1$ for each player i and state x. In a dynamic discrete choice game with

i.i.d. type I extreme value shocks, each player takes the other players' policies as given and computes optimal CCPs using the expected current payoffs and continuation values induced by P. The best-response mapping $\Psi(P, \theta)$ collects these optimal CCPs, and a Markov perfect equilibrium satisfies $P = \Psi(P, \theta)$.

Let $P_0 = \Psi(P_0, \theta_0)$ denote the true equilibrium CCP profile, and let Ψ_j be the approximate best-response mapping used in estimation. For example, replacing the transition matrix F^P induced by a conjectured CCP profile with a numerical approximation F_j^P changes the computed continuation values and hence the best responses. Such errors can propagate further through equilibrium feedback, altering the resulting equilibrium CCPs. As shown below, their effects on parameter estimation depend on how each estimator incorporates the approximate mapping.

To characterize approximation bias across estimators, we use the local metric induced by the multinomial likelihood. For CCP perturbations h and k, define the Fisher (chi-square) inner product

$$\langle h, k \rangle = \mathbb{E}_X \left[\sum_i \sum_{a_i \in A_i} \frac{h_i(a_i \mid X) k_i(a_i \mid X)}{P_{i0}(a_i \mid X)} \right],$$

where P_{i0} denotes player i's equilibrium CCPs. This metric weights deviations by inverse equilibrium choice probabilities, assigning greater weight to less frequent actions. For vector- or matrix-valued perturbations, the inner product is applied column by column, yielding the corresponding matrix of pairwise inner products. In particular, when applied to CCP derivatives, it generates the information matrices entering the bias formulas below.

Let Ψ_P and Ψ_θ denote the Jacobians of Ψ with respect to P and θ, evaluated at (P_0, θ_0). Suppose $I - \Psi_P$ is nonsingular, so that the equilibrium is locally regular. The equilibrium CCP profile $P(\theta)$ is then locally differentiable, with $\partial_\theta P = (I - \Psi_P)^{-1} \Psi_\theta$. Define the one-step approximation residual and the induced equilibrium error as

$$r_j = \Psi_j(P_0, \theta_0) - P_0, \qquad \pi_j = P_j(\theta_0) - P_0 = (I - \Psi_P)^{-1} r_j + O(\delta_{\Psi,j}^2),$$

where $P_j(\theta)$ denotes the local fixed point of $\Psi_j(\cdot, \theta)$ and $\delta_{\Psi,j}$ measures the approximation error in the mapping and its first derivatives. Under suitable smoothness and regularity conditions, the local existence and uniqueness of $P_j(\theta)$ follow for sufficiently large j.

We consider three estimators that differ in how they incorporate the approximate CCP mapping. The two-step (2S) pseudo-likelihood estimator evaluates Ψ_j at a preliminary nonparametric CCP estimate $\hat{P}$, satisfying $\hat{P} \to_p P_0$, and maximizes the resulting pseudo-likelihood (Hotz and Miller, 1993; Aguirregabiria and Mira, 2007). The nested pseudo-likelihood (NPL) estimator instead jointly imposes likelihood optimality and consistency between the CCPs used to evaluate continuation values and those implied by the esti-

mated parameters (Aguirregabiria and Mira, 2007; Kasahara and Shimotsu, 2012). Finally, the full-solution maximum likelihood (ML) estimator solves the approximate equilibrium $P_j(\theta) = \Psi_j(P_j(\theta), \theta)$ for each candidate parameter value and maximizes the resulting likelihood, whether implemented by NFXP or MPEC.

Although all three estimators rely on the same approximate mapping Ψ_j, they differ in how numerical errors enter estimation. To formalize these differences, for a CCP profile $R = \{R_i\}_i$, define the sample pseudo-log-likelihood as $\hat{Q}_T(R) = \frac{1}{T}\sum_{t=1}^{T}\sum_i \log R_i(y_{it} \mid x_t)$, where y_{it} denotes player i's observed action. The two-step (2S), nested pseudo-likelihood (NPL), and full-solution maximum likelihood (ML) estimators are defined by

$$\begin{aligned}
\hat{\theta}_j^{2S} &= \arg\max_{\theta\in\Theta} \hat{Q}_T(\Psi_j(\hat{P}, \theta)), \\
\hat{\theta}_j^{NPL} &= \arg\max_{\theta\in\Theta} \hat{Q}_T(\Psi_j(\hat{P}_j^{NPL}, \theta)), \quad \hat{P}_j^{NPL} = \Psi_j(\hat{P}_j^{NPL}, \hat{\theta}_j^{NPL}), \\
\hat{\theta}_j^{ML} &= \arg\max_{\theta\in\Theta} \hat{Q}_T(P_j(\theta)), \quad P_j(\theta) = \Psi_j(P_j(\theta), \theta),
\end{aligned}$$

where $\hat{P} \to_p P_0$ is a preliminary CCP estimator, and the NPL conditions characterize a locally selected solution. Let θ_j^e, $e \in \{2S, NPL, ML\}$, denote the corresponding population pseudo-true parameters. The following theorem characterizes their first-order approximation biases.

Theorem 3 (Bias geometry for CCP mappings)**.** *Suppose Ψ and Ψ_j are twice continuously differentiable on a neighborhood $\mathcal{U}$ of (P_0, θ_0), with uniformly bounded and Lipschitz continuous second derivatives. Let $\delta_{\Psi,j} = \|\Psi_j - \Psi\|_{C^2(\mathcal{U})} \to 0$. Assume that the state space is finite, P_0 is strictly positive, $I - \Psi_P$ is nonsingular, the relevant population solutions are locally identified and interior, and the matrices appearing below are nonsingular. Then, for $e \in \{2S, NPL, ML\}$, $\theta_j^e - \theta_0 = B_j^e + O(\delta_{\Psi,j}^2)$, where*

$$\begin{aligned}
B_j^{2S} &= -\langle \Psi_\theta, \Psi_\theta \rangle^{-1} \langle \Psi_\theta, r_j \rangle, \\
B_j^{NPL} &= -\langle \Psi_\theta, \partial_\theta P \rangle^{-1} \langle \Psi_\theta, \pi_j \rangle, \\
B_j^{ML} &= -\langle \partial_\theta P, \partial_\theta P \rangle^{-1} \langle \partial_\theta P, \pi_j \rangle.
\end{aligned}$$

Here, Ψ_P and Ψ_θ are evaluated at (P_0, θ_0), $\partial_\theta P = (I - \Psi_P)^{-1}\Psi_\theta$, $r_j = \Psi_j(P_0, \theta_0) - P_0$, and $\pi_j = P_j(\theta_0) - P_0$.

Moreover, replacing Ψ_θ and $\partial_\theta P$ in the bias formulas by their approximate-model counterparts $\Psi_{j,\theta}$ and $\partial_\theta P_j$, evaluated at (P_0, θ_0) and θ_0, respectively, preserves the $O(\delta_{\Psi,j}^2)$ remainder.

The equilibrium derivative and the estimator-specific first-order conditions build on the established CCP estimation framework (Aguirregabiria and Mira, 2007; Kasahara and Shi-

motsu, 2012). Theorem 3 uses these objects to characterize how a common numerical perturbation generates different parameter biases across two-step, NPL, and full-solution estimation. In particular, the estimators differ not only in their sampling properties but also in the numerical errors to which their parameter estimates are locally sensitive.

As in Corollary 1, the exact derivatives in Theorem 3 can be replaced by their approximate-model counterparts $\Psi_{j,\theta}$ and $\partial_\theta P_j$, yielding bias formulas with second-order remainders under suitable approximation-rate conditions. These expressions form the basis of the numerical diagnostics in Section 7. The theorem also reveals several distinctions in how approximation errors affect alternative estimators.

The ML bias B_j^{ML} is determined by the orthogonal projection, under the Fisher metric, of the equilibrium CCP error onto the tangent space of the equilibrium model, paralleling Theorem 2. The NPL bias B_j^{NPL} instead corresponds to an oblique projection of the same equilibrium error, with Ψ_θ determining the projection direction. The matrix $\langle \Psi_\theta, \partial_\theta P \rangle$ corresponds to $\Omega_{\theta\theta} + \Upsilon$ in the notation of Kasahara and Shimotsu (2012). The two-step bias differs because it projects the one-step residual r_j rather than the equilibrium error π_j. Thus, differences across estimators concern not only statistical efficiency but also how numerical errors are propagated and projected onto parameter-sensitive directions. A similar result holds for minimum distance estimation: with weighting matrix $\mathcal{W}$, the two-step bias is $B_j^{MD} = -(\Psi_\theta' \mathcal{W} \Psi_\theta)^{-1} \Psi_\theta' \mathcal{W} r_j$, which coincides with B_j^{2S} under the Fisher weighting matrix.

These distinctions disappear in the single-agent benchmark. As shown by Aguirregabiria and Mira (2002), the policy iteration mapping satisfies $\Psi_P = 0$ at the fixed point. Consequently, $\pi_j = r_j + O(\delta_{\Psi,j}^2)$ and $\partial_\theta P = \Psi_\theta$, so all three estimators have the same first-order approximation bias. The differences in dynamic games therefore arise from strategic equilibrium feedback rather than from the CCP representation itself.

In dynamic games, Ψ_P is generally nonzero, and the equilibrium error is given to first order by $\pi_j = (I - \Psi_P)^{-1} r_j$. The operator $(I - \Psi_P)^{-1}$ determines how numerical errors propagate through strategic interactions before affecting estimation. This propagation depends on the direction of the error. Along an eigenvector of Ψ_P with real eigenvalue λ, the corresponding residual component is multiplied by $(1-\lambda)^{-1}$. Positive eigenvalues below one amplify errors, as may occur under strategic complementarity, whereas negative eigenvalues attenuate them, as under strategic substitution. More generally, the magnitude of the propagated error is bounded by $\|(I - \Psi_P)^{-1}\|$, which can become large near an equilibrium singularity. Even moderate feedback can change the composition of numerical errors across equilibrium directions and, through the bias projections, across structural parameters. There is therefore no general ranking of two-step, NPL, and ML estimators in terms of robustness to numerical approximation. Their relative sensitivity depends on the structure of strategic interaction, the direction of the numerical error, and the parameter of interest. Table 5 in

Section 7 illustrates these differences in the dynamic entry-game application.

The regularity condition $I - \Psi_P$ being nonsingular should be distinguished from the convergence condition for the NPL algorithm. Theorem 3 can apply even when the spectral radius of Ψ_P exceeds one and standard NPL iterations fail to converge (Kasahara and Shimotsu, 2012). In such cases, a locally defined NPL solution may still exist but requires a different numerical method. Finally, a useful benchmark arises when the numerical residual is locally equivalent to a parameter perturbation. If $r_j = \Psi_\theta c$ for some vector c, then $\pi_j = \partial_\theta P\, c$ to first order, and all three bias formulas yield $B_j^e = -c$. Thus, equilibrium feedback changes the CCP representation of the error without changing the implied parameter displacement. Differences across estimators arise from residual components that cannot be represented as local parameter changes, and even substantial amplification of equilibrium error need not generate comparably large parameter bias.

5 Asymptotics with Drifting Approximations

We now allow numerical accuracy to improve with sample size by letting the approximation index depend on T, denoted by $j = j_T$. Write $\delta_T = \delta_{j_T}$, with analogous notation for derivative approximation errors. This framework allows us to assess how rapidly numerical errors must vanish relative to sampling uncertainty. We first establish the asymptotic properties of the estimator around its approximation-specific pseudo-true parameter and then characterize how approximation bias affects inference on θ_0. We also extend the analysis to two-step and NPL estimators and translate the resulting rate conditions into requirements on numerical resolution, highlighting their computational implications in high-dimensional state spaces.

We begin with likelihood-based estimation and impose the following regularity conditions to characterize the estimator's asymptotic behavior as numerical accuracy improves with sample size.

Assumption 3. (i) Assumption 1 holds, and the exact log-likelihood satisfies a uniform law of large numbers on Θ.

(ii) The smoothness and identification conditions in Assumption 2(i)–(ii) hold. Moreover, $\mathbb{E}[\sup_{\theta\in\mathcal{N}} \|\nabla_\theta \log p(Y \mid X;\theta)\|^2] < \infty$, and the exact log-likelihood Hessian satisfies a uniform law of large numbers on a neighborhood of θ_0.

Theorem 4. *The following results hold:*

(i) Under Assumption 3(i), if $\delta_T \to 0$, then $\hat{\theta}_{j_T} \to_p \theta_0$.

(ii) If Assumption 3(ii) also holds and $\delta'_T \to 0$ and $\delta''_T \to 0$, then $\sqrt{T}(\hat{\theta}_{j_T} - \theta_{j_T}) \Rightarrow N(0, I^{-1})$, and $\hat{I}_{j_T} = -\nabla^2_\theta \hat{Q}_{j_T}(\hat{\theta}_{j_T}) \to_p I$.

(iii) Under the conditions in (ii), if $\sqrt{T}\delta_T(\delta_T + \delta'_T + \delta''_T) \to 0$*, then* $\sqrt{T}(\hat{\theta}_{j_T} - \theta_0 - B_{j_T}) \Rightarrow N(0, I^{-1})$*. In particular, if* $\sqrt{T}\delta_T \to 0$*, numerical approximation is asymptotically negligible. If instead* $\sqrt{T}B_{j_T} \to b$*, then* $\sqrt{T}(\hat{\theta}_{j_T} - \theta_0) \Rightarrow N(b, I^{-1})$*. For any component with* $b_k \neq 0$*, the corresponding nominal two-sided Wald confidence interval has asymptotic coverage strictly below its nominal level.*

Part (i) revisits the consistency result of Ackerberg et al. (2009), while part (iii) connects the score-projection representation of Theorem 2 to asymptotic inference. This analysis complements the more general expansions of approximate estimators developed by Kristensen and Salanié (2017) by characterizing the leading bias in terms of the likelihood distortion induced by numerical approximation. For inference, the relevant comparison is between each component of B_j and its standard error $\sqrt{(I^{-1})_{kk}/T}$, rather than the accumulated discrepancy in the joint likelihood. For a fixed approximation, the latter satisfies $\sum_{t=1}^{T} g_j(W_t) = T\mathbb{E}[g_j(W)] + O_p(\sqrt{T}\delta_j)$ and can grow with sample size, as emphasized by Fernández-Villaverde et al. (2006). By contrast, B_j is a population displacement that does not itself grow with T. Likelihood normalization implies $\mathbb{E}[g_j(W)] = O(\delta_j^2)$, while the local slope of the population criterion distortion satisfies $\nabla_\theta(Q_j - Q)(\theta_0) = -\mathbb{E}[s(W)g_j(W)] + O(\delta_j^2 + \delta_j\delta'_j)$. It is this local distortion, rather than the accumulated likelihood discrepancy, that determines the leading parameter bias.

For two-step and NPL estimators, Aguirregabiria and Mira (2007) provide the benchmark asymptotic distributions under the exact equilibrium mapping, while Kasahara and Shimotsu (2012) further analyze the convergence properties of NPL and related sequential estimators. Under conditions ensuring that these asymptotic expansions remain valid as numerical accuracy improves, the limiting sampling distributions centered at the approximation-specific pseudo-true parameters retain their exact-model counterparts. Combined with Theorem 3, numerical approximation then introduces an estimator-specific location shift $\sqrt{T}B^e_{j_T}$, $e \in \{2S, NPL\}$, provided the higher-order approximation remainder is negligible at the root-T scale. Thus, numerical approximation affects the centering of the limiting distribution, while first-step estimation and equilibrium feedback continue to determine its variance. The score-projection and fixed-point representations identify the direction and magnitude of these shifts for each estimator.

These asymptotic requirements can translate into substantial computational costs. Suppose the state space has d continuous dimensions and the approximation error satisfies $\delta_j \asymp h^p$, where h denotes the mesh size and p the approximation order. For example, $p = 1$ may arise with piecewise-constant discretization of a Lipschitz function, while $p = 2$ is attainable with piecewise-linear interpolation or midpoint quadrature under suitable smoothness conditions. With n grid points, $h \asymp n^{-1/d}$, so the conventional negligibility condition $\sqrt{T}\delta_T \to 0$ requires $n \gg T^{d/(2p)}$. For $d = 3$, $p = 1$, and $T = 10^4$, the corresponding bench-

mark is 10^6 grid points, with asymptotic negligibility requiring substantially finer resolution. Repeatedly solving the model at this accuracy can be computationally demanding even in moderately high-dimensional state spaces. The bias corrections developed in the next section exploit the explicit characterization of approximation bias to relax the accuracy required during optimization, while using a more accurate model solution only at the final estimate.

6 Bias Correction and Numerical Diagnostics

The asymptotic requirements of Section 5 can make accurate structural estimation computationally demanding, particularly when solving the model is costly and must be repeated throughout optimization. A natural alternative, developed in a general setting by Kristensen and Salanié (2017), is to estimate the model using a coarse approximation and apply a local adjustment based on a more accurate solution. Our fixed-point analysis provides additional structure for this approach: it identifies the numerical errors relevant for parameter estimation and allows their leading effect to be computed directly from discrepancies in the implied choice probabilities.

Let $j^\star$ index a reference approximation that is sufficiently accurate for inference but too costly to use throughout optimization. Starting from the coarse estimator $\hat{\theta}_j$, we consider two corrections. The first uses the score-projection representation to estimate and remove the leading approximation bias. The second takes a Newton-type step toward the reference-model estimator, following Kristensen and Salanié (2017). Both require accurate model evaluations only locally, rather than repeated optimization using the reference approximation. The resulting bias estimates also provide a diagnostic of numerical accuracy relative to sampling uncertainty.

The first correction follows from Corollary 1. For discrete choice models, compute the discrepancy between the coarse and reference CCPs at $\hat{\theta}_j$: $\hat{\pi}_j(a \mid x) = P_j(a \mid x; \hat{\theta}_j) - P_{j^\star}(a \mid x; \hat{\theta}_j)$. The plug-in bias-corrected estimator is

$$\hat{\theta}_j^{BC} = \hat{\theta}_j - \hat{B}_j, \qquad \hat{B}_j = -\hat{I}_j^{-1} \frac{1}{T} \sum_{t=1}^{T} \sum_{a \in A} \frac{\nabla_\theta P_j(a \mid x_t; \hat{\theta}_j)}{P_j(a \mid x_t; \hat{\theta}_j)} \hat{\pi}_j(a \mid x_t),$$

where $\hat{I}_j = -\nabla_\theta^2 \hat{Q}_j(\hat{\theta}_j)$. For continuous outcomes, the sum over actions is replaced by the corresponding integral. The correction uses derivatives of the coarse model and requires the reference model to be solved only at $\hat{\theta}_j$. Rather than attempting to reduce the overall discrepancy between the two likelihoods, it removes the component of the local probability distortion that contributes to parameter bias. In contrast to generic analytical corrections based on an explicit expansion of the approximation scheme, this construction applies directly

to the discrepancy between the approximate and reference probability models.

An alternative is to update the coarse estimate using the score of the reference likelihood:

$$\hat{\theta}_j^{NR} = \hat{\theta}_j + \hat{I}_j^{-1} \nabla_\theta \hat{Q}_{j^\star}(\hat{\theta}_j).$$

This is a one-step Newton-type correction using the coarse-model information matrix as an approximation to the reference curvature. It is closely related to the Newton–Raphson adjustment of Kristensen and Salanié (2017), which updates a preliminary estimator using a more accurate approximation of the objective. When the reference score is computed by central finite differences, the update typically requires $2k$ reference-model evaluations for $k = \dim \theta$, in addition to any evaluations needed to construct the baseline solution. Its advantage is that it does not require an explicit projection of the probability discrepancy, although it generally involves more reference evaluations than the plug-in correction.

The two corrections exploit the same local information but differ in how it is obtained. The Newton-type correction evaluates the reference-model score at the coarse estimate, whereas our plug-in correction uses only the discrepancy between coarse and reference choice probabilities, together with derivatives already available from the coarse model. By likelihood normalization, the plug-in adjustment can also be interpreted as replacing the realized difference between reference and coarse scores with its conditional expectation under the reference model. This avoids differentiating the computationally expensive reference solution and, when reference scores require numerical differentiation, can substantially reduce the number of accurate model evaluations.

The plug-in approach offers two additional advantages. First, it does not require knowledge of the numerical approximation scheme or its convergence order; it applies directly to probability discrepancies generated by discretization, interpolation, or other approximations that preserve likelihood normalization. Second, the same estimated bias provides a parameter-specific diagnostic of numerical accuracy, allowing researchers to assess whether correction is needed before applying it. Thus, while both methods achieve second-order accuracy under suitable conditions, the plug-in correction provides a probability-based implementation and diagnostic tailored to structural likelihood models. Neither procedure uniformly dominates the other, but the plug-in approach is particularly attractive when accurate probability evaluations are available and reference-model derivatives are costly.

The following theorem establishes their second-order accuracy relative to the coarse approximation.

Theorem 5 (Bias corrections for maximum likelihood)**.** *Consider the approximate maximum likelihood estimator $\hat{\theta}_j$ defined in Section 2, including full-solution ML in dynamic games. Let $j = j_T$ and $j^\star = j_T^\star$ denote the coarse and reference approximations, and write $\kappa_j = \delta_j +$*

$\delta'_j + \delta''_j$. *Suppose Assumption 3 and the conditions of Corollary 1 hold, with* $\kappa_j \to 0$. *Assume that* $\delta_{j^\star} \le \delta_j$ *and* $\delta^{(k)}_{j^\star} \le \delta^{(k)}_j$ *for* $k = 1, 2$. *Suppose further that: (a) The sample Hessians are locally Lipschitz on* $\mathcal{N}$ *with* $O_p(1)$ *Lipschitz constants, and* $\mathbb{E}[\|\nabla^2_\theta \log p(Y \mid X; \theta_0)\|^2] < \infty$; *(b) The empirical CCP-projection term* $F_{T,j}(\theta) = \frac{1}{T}\sum_{t=1}^{T}\sum_{a\in A} \nabla_\theta \log P_j(a \mid X_t; \theta)\{P_j - P_{j^\star}\}(a \mid X_t; \theta)$ *is locally Lipschitz on* $\mathcal{N}$ *with stochastic modulus* $O_p(\delta_j + \delta'_j)$. *Then the following results hold:*

(i) The Newton-type correction satisfies $\hat\theta^{NR}_j - \hat\theta_{j^\star} = O_p(\delta_j \kappa_j) + o_p(T^{-1/2})$.

(ii) The plug-in correction satisfies $\hat\theta^{BC}_j - \theta_0 - (\hat\theta_j - \theta_j) = O_p\left(\delta_j\kappa_j + \delta_{j^\star} + (\delta_j + \delta'_j)T^{-1/2}\right)$.

Consequently, if $\sqrt{T}\delta_j\kappa_j \to 0$ *and* $\sqrt{T}\delta_{j^\star} \to 0$, *both corrected estimators satisfy*

$$\sqrt{T}(\hat\theta^e_j - \theta_0) \Rightarrow N(0, I^{-1}), \qquad e \in \{BC, NR\}.$$

Theorem 5 separates the remaining numerical error into two components: the higher-order error associated with coarse optimization and the approximation error of the reference solution. When level and derivative approximation errors have comparable rates, the coarse approximation need only satisfy $\sqrt{T}\delta_j^2 \to 0$, rather than the conventional requirement $\sqrt{T}\delta_j \to 0$. In the grid setting of Section 5, where $\delta_j \asymp n^{-p/d}$, this relaxes the resolution requirement inside the optimizer from $n \gg T^{d/(2p)}$ to $n \gg T^{d/(4p)}$. The reference approximation must still satisfy the original accuracy requirement, but it is evaluated only a limited number of times. The gain is therefore computational rather than a relaxation of the final accuracy needed for inference.

Theorem 5 applies to likelihood-based estimation, including full-solution ML. For two-step and NPL estimators, the corresponding corrections require estimator-specific bias representations and sampling distributions. The following theorem establishes these results under additional regularity conditions.

Assumption 4 (Asymptotic linearity along the approximation sequence)**.** (i) The first-step estimator satisfies $\|\hat P - P_0\| = O_p(a_T)$ with $a_T \to 0$. (ii) For $e \in \{2S, NPL\}$, the locally selected estimators admit the expansion $\sqrt{T}(\hat\theta^e_j - \theta^e_j) = T^{-1/2}\sum_{t=1}^{T}\varphi_e(W_t) + o_p(1)$, uniformly along the approximation sequence $j = j_T$, where φ_e is the mean-zero influence function of the corresponding exact-model estimator and $V_e = \mathbb{E}[\varphi_e(W)\varphi_e(W)']$ is finite and nonsingular.

Assumption 4(ii) is the exact-model asymptotic linearity established by Aguirregabiria and Mira (2007) for the two-step and NPL estimators, and by Kasahara and Shimotsu (2012) for locally selected NPL solutions, applied along a sequence of approximate mappings Ψ_{j_T} with $\|\Psi_{j_T} - \Psi\|_{C^2} \to 0$. Because the approximate mapping converges to the exact one in C^2, the regularity conditions underlying those results hold uniformly along the sequence, and the

influence function is that of the exact model. We take this as a maintained hypothesis rather than reproducing the argument; the contribution of the theorem is the characterization and removal of the centering term, not the sampling distribution.

Theorem 6 (Bias correction for two-step and NPL under asymptotic linearity)**.** *Maintain the conditions of Theorem 3, with finite state and action spaces and CCPs bounded away from zero locally. Let $j = j_T$ and $j^\star = j_T^\star$ denote coarse and reference approximations, with $d_j = \delta_{\Psi,j} \to 0$ and $d_{j^\star} \leq d_j$. For CCP profiles, define the empirical Fisher inner product*

$$\langle h, k \rangle_{T,R} = \frac{1}{T} \sum_{t=1}^{T} \sum_{i} \sum_{a_i \in A_i} \frac{h_i(a_i \mid X_t) k_i(a_i \mid X_t)}{R_i(a_i \mid X_t)}.$$

For two-step estimation, let $R_j^{2S} = \Psi_j(\hat{P}, \hat{\theta}_j^{2S})$, $A_j^{2S} = \Psi_{j,\theta}(\hat{P}, \hat{\theta}_j^{2S})$, and $\hat{r}_j = R_j^{2S} - \Psi_{j^\star}(\hat{P}, \hat{\theta}_j^{2S})$. For NPL, let $R_j^{NPL} = P_j(\hat{\theta}_j^{NPL})$, $A_j^{NPL} = \Psi_{j,\theta}(R_j^{NPL}, \hat{\theta}_j^{NPL})$, $D_j^{NPL} = (I - \Psi_{j,P})^{-1} A_j^{NPL}$, where $\Psi_{j,P}$ is evaluated at $(R_j^{NPL}, \hat{\theta}_j^{NPL})$, and $\hat{\pi}_j = P_j(\hat{\theta}_j^{NPL}) - P_{j^\star}(\hat{\theta}_j^{NPL})$. Define the feasible bias corrections by

$$\hat{B}_j^{2S} = -\langle A_j^{2S}, A_j^{2S} \rangle_{T,R_j^{2S}}^{-1} \langle A_j^{2S}, \hat{r}_j \rangle_{T,R_j^{2S}}, \quad \hat{B}_j^{NPL} = -\langle A_j^{NPL}, D_j^{NPL} \rangle_{T,R_j^{NPL}}^{-1} \langle A_j^{NPL}, \hat{\pi}_j \rangle_{T,R_j^{NPL}}. \tag{4}$$

Further suppose Assumption 4 holds. Then $\hat{B}_j^e - B_j^e = O_p(d_j^2 + d_{j^\star} + d_j(a_T + T^{-1/2}))$. Consequently, if $\sqrt{T}\, d_j^2 \to 0$, $\sqrt{T}\, d_{j^\star} \to 0$, and $\sqrt{T}\, d_j a_T \to 0$, the corrected estimators $\hat{\theta}_j^{e,BC} = \hat{\theta}_j^e - \hat{B}_j^e$ satisfy $\sqrt{T}(\hat{\theta}_j^{e,BC} - \theta_0) \Rightarrow N(0, V_e)$ for $e \in \{2S, NPL\}$.

Theorem 6 extends the second-order correction principle to two-step and NPL estimation. Under $\sqrt{T}$ first-step accuracy and a sufficiently accurate reference approximation, the coarse mapping need only satisfy $\sqrt{T} d_j^2 \to 0$. The correction removes the leading estimator-specific numerical bias without altering the first-order sampling distribution. Thus, the exact-model variance formulas of Aguirregabiria and Mira (2007) and Kasahara and Shimotsu (2012), including first-step estimation effects where relevant, remain applicable to the corrected estimators.

A related approach is Richardson extrapolation, which removes the leading approximation error by combining estimates computed at different resolutions. If $\theta_h - \theta_0 = ch^p + o(h^p)$ for a known order p, then $(2^p \hat{\theta}_{h/2} - \hat{\theta}_h)/(2^p - 1)$ eliminates the leading h^p term. This approach can be effective when the approximation admits a stable expansion, but such regularity may be difficult to establish in structural fixed-point problems. Boundary effects, discrete state components, and changes in interpolation error can alter the dominant approximation term across resolutions. The numerical experiments in Section 7 illustrate this difficulty. Our reference-model corrections do not require knowledge of the approximation order or an explicit expansion in grid resolution, making the two approaches complementary.

Beyond bias correction, the projection formula provides a parameter-specific diagnostic of numerical accuracy. After estimating the model with the coarse approximation, the researcher can evaluate $\hat{B}_j$ using a reference solution at $\hat{\theta}_j$ and compare each component with its conventional standard error. For parameter ℓ, the ratio $|\hat{B}_{j,\ell}|/\operatorname{se}(\hat{\theta}_{j,\ell})$ measures the estimated numerical displacement relative to sampling uncertainty. Unlike a discrepancy in the maximized likelihood or a sup-norm value-function error, this diagnostic directly assesses the numerical error relevant for inference on the parameter of interest.

When the estimated bias is substantial relative to its standard error, either correction can be applied, with the adequacy of the reference approximation assessed by repeating the diagnostic at a finer resolution. Conversely, a small estimated bias may indicate that the coarse approximation is already adequate for parameter inference, even when other numerical diagnostics reveal visible discrepancies. This approach complements the approximation-robust inference of Deaner (2020), which accounts for numerical approximation through error bounds and set-based inference. Here, the objective is instead to estimate and remove the leading parameter displacement. Numerical resolution is consequently assessed in terms of its implications for statistical inference, rather than solely through the accuracy of the computed structural solution.

7 Numerical Evidence on Approximation Bias

This section examines the quantitative implications of our theoretical results in two settings. We first consider a continuous-state version of Rust's engine-replacement model. This design allows us to assess the accuracy of the first-order bias representation, distinguish value-function errors from those relevant for estimation, examine the role of discounting and mixing, and evaluate the finite-sample performance of the bias corrections. Population quantities can be computed accurately without simulation, allowing us to isolate numerical approximation error from sampling variability.

We then consider a two-firm dynamic entry game to investigate how equilibrium feedback affects approximation bias across estimators. Holding the approximate CCP mapping fixed, we compare two-step, NPL, and full-solution ML estimation as the strength and sign of strategic interaction vary. The experiments illustrate three distinct mechanisms: the magnitude of primitive numerical errors, their propagation through the fixed-point structure, and their projection onto parameter-sensitive directions. We also examine whether the estimator-specific bias formulas provide useful corrections and diagnostics in finite samples.

7.1 Engine-replacement design

We consider a continuous-state version of the engine-replacement model of Rust (1987). The state is scaled mileage, $x \in [0, 1]$, and the agent chooses between keeping the engine ($a = 0$) and replacing it ($a = 1$). Flow utilities are $u(x, 0) = -\theta_1 x$ and $u(x, 1) = -\theta_2$, with independent type I extreme-value shocks and true parameter $\theta_0 = (4, 6)$. Under keeping, the state evolves according to $x' = \min\{x + \eta, 1\}$, whereas replacement resets mileage so that $x' = \min\{\eta, 1\}$, where $\eta \sim \text{Exp}(10)$.

The numerical approximation partitions $[0, 1]$ into n equal bins, uses exact transition probabilities across bins, and evaluates the integrated value function at bin midpoints. At an observed off-grid state, the approximate continuation value is obtained by one application of the discretized operator in (3). We use the same scheme with $n = 1000$ as a high-accuracy reference model, both to generate the data and to construct the bias corrections. The population distribution of observed states is uniform over 2,000 equally spaced points, so expectations can be evaluated by direct summation. Pseudo-true parameters are obtained by optimizing the population criterion, and their first-order biases are computed using (2).

Table 1 compares actual pseudo-true bias with its first-order prediction for $\beta = 0.95$. The projection formula captures both the direction and magnitude of the numerical displacement, with accuracy improving rapidly as the grid is refined. At $n = 20$, for example, the prediction error is approximately one percent of the actual bias. The remaining discrepancy is consistent with the higher-order remainder in Theorem 2. Thus, the bias representation is informative even when the numerical approximation is relatively coarse.

The final columns report the population version of the one-step Newton correction, $\theta_j + I_j^{-1} \nabla_\theta Q_{j^\star}(\theta_j)$, where $j^\star$ denotes the reference approximation. As anticipated by the local correction argument of Kristensen and Salanié (2017) and Theorem 5, the adjustment removes most of the leading displacement. For $n \leq 40$, the norm of the remaining bias is approximately 0.14–0.22 times $\|B_j\|^2$. At $n = 10$, for example, the bias in θ_2 falls from approximately -0.067 to -0.0007. The exercise illustrates the computational advantage of using a more accurate model only for a local adjustment rather than throughout optimization.

The pattern across grid sizes also cautions against relying exclusively on a prespecified approximation order. Although the overall bias declines substantially with refinement, it is not monotone. At finer resolutions, the boundary atom at $x = 1$ contributes an error of a different order from the interior midpoint approximation, altering the dominant error component. Consequently, the same nominal grid-refinement rule need not deliver a stable reduction in parameter bias. The reference-model correction does not require knowledge of the leading discretization term or its convergence order.

We next examine how discounting affects numerical accuracy. Table 2 varies β from

Table 1: Actual, Predicted, and Newton-Corrected Bias ($\beta = 0.95$)

	Actual bias		Predicted bias		After Newton step	
n	θ_1	θ_2	θ_1	θ_2	θ_1	θ_2
5	-1.74×10^{-1}	-3.01×10^{-1}	-1.62×10^{-1}	-2.63×10^{-1}	-8.0×10^{-3}	-1.5×10^{-2}
10	-3.95×10^{-2}	-6.66×10^{-2}	-3.88×10^{-2}	-6.43×10^{-2}	-4.0×10^{-4}	-7.4×10^{-4}
20	-7.51×10^{-3}	-1.39×10^{-2}	-7.47×10^{-3}	-1.37×10^{-2}	-1.6×10^{-5}	-3.3×10^{-5}
40	-7.09×10^{-4}	-2.21×10^{-3}	-7.06×10^{-4}	-2.20×10^{-3}	-4.7×10^{-7}	-1.1×10^{-6}
80	3.73×10^{-4}	3.19×10^{-5}	3.74×10^{-4}	3.27×10^{-5}	-8.1×10^{-8}	-9.1×10^{-8}
160	3.34×10^{-4}	2.67×10^{-4}	3.34×10^{-4}	2.67×10^{-4}	-3.0×10^{-8}	-2.8×10^{-8}

Notes: $\theta_0 = (4, 6)$. Actual and predicted biases are $\theta_j - \theta_0$ and B_j, respectively. The final columns report $\theta_j + I_j^{-1} \nabla_\theta Q_{j^\star}(\theta_j) - \theta_0$, using the $n = 1000$ reference model. Population expectations are evaluated over 2,000 state-support points.

0.9 to 0.999 while holding the discretization fixed. The sup-norm error in the choice-specific expected value function grows substantially with the discount factor, as suggested by conventional contraction-based bounds. For $n = 10$, it increases from 0.286 to 17.41. In contrast, the within-state oscillation $\mathrm{osc}(e_j)$ rises only from 0.181 to 0.205. The corresponding parameter biases change much less dramatically than the sup-norm error, and their first-order predictions remain accurate throughout the experiment.

This difference reflects the role of relative continuation values. With logit choice probabilities, components of approximation error common to all actions at a given state cancel from the likelihood. Proposition 2 therefore bounds the likelihood distortion using $\mathrm{osc}(e_j)$ rather than the overall level of value-function error. The replacement structure also generates mixing in the controlled Markov process, limiting the persistence of relative errors. As a result, approximation error relevant for estimation need not increase at the conventional rate $(1-\beta)^{-1}$. This interpretation is consistent with the importance of value differences and transition variation emphasized by Deaner (2020).

Table 3 examines the constants underlying the span bound in Proposition 2(c). The realized ratio $\mathrm{osc}(e_j)/\,\mathrm{sp}(r_j)$ remains between 0.88 and 0.91 as β increases. By Remark 1, the Dobrushin sum $\bar{C}_j = \sum_{k \geq 0} \beta^k \tau(\bar{M}_j^k)$ can be bounded using the restriction of $\bar{M}_j$ to the grid. With $n = 10$, the resulting 20-state matrix T_j yields bounds $C_j^G \leq \bar{C}_j \leq \bar{C}_j^+ = 1 + \beta C_j^G$ that apply to the entire continuous state space. The reported upper bound increases only from 3.80 to 4.45 as β rises from 0.9 to 0.999, while the lower bound ranges from 3.12 to 3.46. The series tail is bounded below 10^{-12}. The resulting scaled Dobrushin values are approximately four to five times the realized errors, remaining substantially more informative than the Doeblin values. Two inputs to the scaled bounds are themselves numerical: $\bar{P}_j$ is constructed from the 1,000-bin reference solution in place of the exact value function, and $\mathrm{sp}(r_j)$ is evaluated on 2,000 state points. The constant $\bar{C}_j^+$ is exact over the continuum given $\bar{P}_j$; the scaled bounds inherit these two approximations.

Table 2: Discounting, Value-Function Error, and Parameter Bias

					Actual bias		Predicted
n	β	$(1-\beta)^{-1}$	$\sup\lvert e_j\rvert$	$\operatorname{osc}(e_j)$	θ_1	θ_2	$B_{j,2}$
10	0.900	10	0.286	0.181	−0.0310	−0.0634	−0.0615
10	0.950	20	0.473	0.193	−0.0395	−0.0666	−0.0643
10	0.990	100	1.865	0.203	−0.0464	−0.0679	−0.0651
10	0.999	1000	17.41	0.205	−0.0479	−0.0680	−0.0650
20	0.900	10	0.076	0.049	−0.0050	−0.0128	−0.0126
20	0.950	20	0.124	0.052	−0.0075	−0.0139	−0.0137
20	0.990	100	0.477	0.056	−0.0095	−0.0144	−0.0142
20	0.999	1000	4.418	0.056	−0.0099	−0.0144	−0.0143

Notes: e_j is the choice-specific expected value error relative to the $n = 1000$ reference model, evaluated at observed states. The within-state oscillation is $\operatorname{osc}(e_j) = \sup_x |e_j(x,0) - e_j(x,1)|$. $B_{j,2}$ is the first-order prediction for the bias in θ_2.

Table 3: Dobrushin and Doeblin Bounds ($n = 10$)

			Realized	Bound constant		Scaled bound	
β	$\operatorname{sp}(r_j)$	$\operatorname{osc}(e_j)$	Ratio	$\bar{C}_j^+$	Doeblin	$\bar{C}_j^+$	Doeblin
0.900	0.199	0.181	0.91	3.80	10.3	0.76	2.05
0.950	0.217	0.193	0.89	4.12	19.5	0.89	4.22
0.990	0.231	0.203	0.88	4.39	78.5	1.01	18.1
0.999	0.234	0.205	0.88	4.45	258.6	1.04	60.6

Notes: The realized ratio is $\operatorname{osc}(e_j)/\operatorname{sp}(r_j)$. The Dobrushin upper bound $\bar{C}_j^+ = 1 + \beta C_j^G$ is computed from the 20-state grid kernel (Remark 1), using a 1,000-bin reference solution. The series is truncated at $k = 80$, with a geometric tail bound below 10^{-12}. The Doeblin benchmark uses $\rho_j \approx 0.005$. The constant $\bar{C}_j^+$ is uniform over the continuous state space by Remark 1. The scaled bounds multiply each constant by $\operatorname{sp}(r_j)$, which is evaluated on 2,000 state points, and $\bar{P}_j$ is constructed from the 1,000-bin reference solution in place of the exact value function; the scaled bounds are therefore subject to these two sources of numerical approximation. For $n = 20$, $\bar{C}_j^+$ ranges from 3.93 to 4.67.

The two-step Doeblin bound is considerably more conservative. The minorization probability for $\bar{M}_j$ is $\rho_j = \min_{x \in G} \bar{P}_j(1 \mid x) \approx 0.005$, reflecting the relatively low replacement probability. The corresponding multiplier $2/\{1 - \beta^2(1 - \rho_j)\}$ increases from approximately 10 to 259. At $\beta = 0.999$, the scaled Doeblin value is about 61, compared with the realized error of 0.205. Thus, although the Doeblin condition provides a discount-factor-uniform multiplier when ρ_j is uniformly bounded away from zero, its numerical bound is too conservative for guiding numerical accuracy choices.

These results distinguish the overall value-function error from the relative error relevant for estimation. The computable Dobrushin upper bound $\bar{C}_j^+$ provides a more informative measure of relative-value error than the conservative Doeblin bound. Its construction uses integrated CCPs obtained from a reference solution; using approximate CCPs instead provides a simpler diagnostic but requires additional error control to preserve the bound.

We finally examine the inferential consequences of numerical approximation. Table 4 reports 300 Monte Carlo replications for $T \in \{10^3, 10^4, 10^5\}$, comparing coarse NFXP estimation with the plug-in bias correction and the one-step Newton adjustment. Standard errors are computed from the coarse-model information matrix.

The results illustrate why approximation bias must be assessed relative to sampling uncertainty. At a fixed grid resolution, the coarse estimator's mean bias changes little with T. In contrast, the discrepancy in the joint log-likelihood grows with sample size: for $n = 5$, the mean absolute discrepancy at θ_0 increases from approximately 1.1 to 6.4 and then to 62 as T rises. The growing likelihood discrepancy does not imply that the population parameter displacement is increasing. Instead, the same numerical bias becomes more consequential because sampling uncertainty declines, consistent with the distinction between likelihood error and parameter displacement emphasized by Fernández-Villaverde et al. (2006); Ackerberg et al. (2009).

This mechanism produces substantial coverage deterioration. For $n = 5$, coverage of nominal 95% intervals falls from approximately 87% at $T = 10^3$ to 41–45% at $T = 10^4$, and to zero at $T = 10^5$. With $n = 10$, coverage remains relatively close to nominal at the smaller sample sizes but declines to approximately 70% at $T = 10^5$. Thus, a numerical approximation that appears adequate in a moderate sample can become inadequate for inference as statistical precision increases.

Both corrections substantially reduce the bias and improve coverage. For $n = 10$, the corrected estimators achieve coverage near nominal levels even at $T = 10^5$. With the coarser $n = 5$ approximation, corrected coverage remains around 90–92% in the largest sample. This residual undercoverage is consistent with higher-order approximation error becoming detectable as sampling uncertainty declines. Overall, the simulations illustrate the practical gain from removing the parameter-relevant component of numerical error, rather than requiring uniformly accurate structural solutions throughout optimization.

7.2 Dynamic entry game

We next examine the estimator-specific propagation of numerical errors in a symmetric two-firm dynamic entry game, following the framework of Aguirregabiria and Mira (2007). The state is (x, k_1, k_2), where $x \in [0, 1]$ denotes market size and k_i indicates whether firm i was active in the previous period. An active firm receives flow payoff $\theta_{RS}x - \theta_{RN}a_{-i} - \theta_{EC}(1 - k_i) - 1$ plus a type I extreme-value shock, while an inactive firm receives zero. Market size follows $x' = \min\{\max\{0.8x + 0.1 + 0.15\varepsilon', 0\}, 1\}$, with $\varepsilon' \sim N(0, 1)$. We set $\beta = 0.95$, $\theta_{RS} = 3$, and $\theta_{EC} = 1.5$, and vary θ_{RN} from -3 to 4. Negative values of θ_{RN} generate strategic complementarity, whereas positive values generate strategic substitutability.

The coarse approximation discretizes market size into $n = 6$ bins, and the reference model

Table 4: Engine Replacement: Monte Carlo Bias, RMSE, and 95% Coverage

			Mean bias		RMSE		Coverage	
n	T	Estimator	θ_1	θ_2	θ_1	θ_2	θ_1	θ_2
5	10^3	Coarse NFXP	-0.160	-0.263	0.303	0.538	0.873	0.867
		Plug-in BC	0.011	0.044	0.275	0.485	0.923	0.937
		One-step NR	-0.002	0.019	0.272	0.478	0.917	0.943
5	10^4	Coarse NFXP	-0.179	-0.301	0.196	0.336	0.410	0.447
		Plug-in BC	-0.009	0.006	0.086	0.154	0.947	0.947
		One-step NR	-0.021	-0.018	0.087	0.152	0.923	0.933
5	10^5	Coarse NFXP	-0.172	-0.297	0.175	0.301	0.000	0.000
		Plug-in BC	-0.003	0.010	0.030	0.052	0.913	0.920
		One-step NR	-0.014	-0.014	0.033	0.052	0.900	0.920
10	10^3	Coarse NFXP	-0.040	-0.074	0.275	0.479	0.933	0.927
		Plug-in BC	-0.001	-0.007	0.277	0.476	0.953	0.933
		One-step NR	-0.001	-0.008	0.277	0.476	0.950	0.933
10	10^4	Coarse NFXP	-0.042	-0.064	0.094	0.165	0.923	0.923
		Plug-in BC	-0.003	0.003	0.086	0.153	0.957	0.957
		One-step NR	-0.004	0.002	0.086	0.152	0.953	0.953
10	10^5	Coarse NFXP	-0.038	-0.066	0.047	0.082	0.690	0.707
		Plug-in BC	0.001	0.001	0.029	0.049	0.957	0.940
		One-step NR	0.000	-0.000	0.029	0.049	0.953	0.937

Notes: 300 replications, $\beta = 0.95$, and $\theta_0 = (4, 6)$. Data are generated using the $n = 1000$ reference model, which is also used for the corrections. Coverage is based on Wald intervals using the coarse-model information matrix. The Monte Carlo standard error of a coverage probability near 0.95 is approximately 0.013.

uses 120 bins. At each observed state, CCPs are obtained by solving the local equilibrium conditional on the approximate continuation values. Thus, numerical error enters through the continuation-value approximation without introducing a separate interpolation rule for equilibrium CCPs. The population distribution is uniform over 100 equally spaced market-size points and the four incumbency configurations.

For each strategic regime, we compute the pseudo-true parameters of three estimators: two-step estimation with the true CCP profile supplied in the first step, NPL obtained from its fixed-point and first-order conditions, and full-solution ML. The same approximate best-response mapping Ψ_j is used throughout. Consequently, differences in approximation bias are attributable to the estimator-specific projection and equilibrium-feedback mechanisms in Theorem 3, rather than to differences in numerical approximation schemes.

Table 5 reports the resulting population biases. Panel A presents the actual displacement $\theta_j^e - \theta_0$ for each estimator $e \in \{2S, NPL, ML\}$. Panel B reports the remaining bias after applying the feasible first-order correction $\tilde{B}_j^e$ implied by Theorem 3. The correction uses derivatives of the coarse model and the discrepancy between coarse and reference solutions,

evaluated at each estimator's own pseudo-true parameter.

The first finding is that the bias representations are quantitatively accurate. Evaluating the theoretical first-order formulas at θ_0, the largest discrepancy from the actual pseudo-true bias is approximately 0.0018 across the designs. The formulas therefore predict not only the direction of numerical bias but also its magnitude. Panel B provides a more demanding assessment by evaluating the feasible correction at a displaced parameter value, as would be required in empirical implementation.

Second, the three estimators have essentially identical first-order biases when $\theta_{RN} = 0$. In this benchmark, strategic feedback vanishes, so the distinction between one-step and propagated equilibrium errors disappears. This is consistent with Theorem 3, under which two-step, NPL, and ML share the same first-order bias when $\Psi_P = 0$.

Third, strategic interaction produces substantial differences across estimators and parameters, even when equilibrium-error amplification is moderate. The ratio $\|\pi_j\|/\|r_j\|$, measured in the Fisher norm, ranges from 0.74 to 1.56. It exceeds one in the complementarity designs and falls below one in the substitutability designs. Yet these relatively modest changes in the norm of equilibrium error can produce considerably larger changes in individual parameter biases.

Under strategic complementarity, for example, full-solution ML generally exhibits larger biases in the market-size and competition parameters than two-step estimation. At $\theta_{RN} = -2$, the ML market-size bias is approximately 30% larger. Under strategic substitutability, the pattern changes. At $\theta_{RN} = 2$, the ML market-size bias is approximately half the two-step bias, while its entry-cost bias is larger in magnitude, although from a small base. At $\theta_{RN} = 4$, two-step estimation produces the largest bias in the competition parameter, approximately -0.051, compared with -0.022 for ML. NPL often lies between the other estimators, but this is not a general ordering. These results illustrate the distinct projection geometries characterized in Theorem 3 and show why no estimator can be ranked uniformly by robustness to numerical approximation.

Fourth, the feasible correction removes most of the bias across all three estimators. In Panel B, the largest remaining component is approximately 0.0017, compared with uncorrected biases reaching 0.051. In Euclidean norm, the correction eliminates at least 93% of the bias in the reported designs, with a median reduction of approximately 99%. The remaining error is largest at $\theta_{RN} = 4$, where the primitive numerical distortion is also largest. The effectiveness of the correction across estimators is important: although the same approximate mapping generates different biases under alternative estimation methods, their leading effects can be estimated using local reference-model calculations tailored to each estimator.

The population analysis establishes that equilibrium feedback affects numerical bias differently across estimators, but it does not establish whether the resulting displacement is

Table 5: Dynamic Entry Game: Approximation Bias by Estimator

	Eigenvalues of Ψ_P			Two-step			NPL			ML		
θ_{RN}	Max	Min	$\|\pi_j\|/\|r_j\|$	RS	RN	EC	RS	RN	EC	RS	RN	EC
Panel A: pseudo-true bias (×100)												
-3	0.49	-0.21	1.19	0.16	0.08	-0.01	0.15	0.07	-0.03	0.13	0.05	-0.05
-2	0.49	-0.36	1.56	1.48	0.83	0.05	1.62	0.88	0.16	1.92	1.01	0.25
-1	0.32	-0.26	1.42	4.66	2.96	0.00	4.85	2.98	0.13	5.45	3.31	0.28
0	0.00	0.00	1.00	2.81	2.17	0.01	2.82	2.18	0.00	2.81	2.16	0.00
1	0.24	-0.38	0.84	1.44	1.22	-0.02	1.14	0.97	-0.11	0.81	0.52	-0.17
2	0.44	-0.90	0.75	0.64	0.42	-0.04	0.42	0.19	-0.12	0.34	0.06	-0.23
3	0.53	-1.82	0.74	0.39	-0.39	-0.16	0.48	-0.14	-0.30	0.28	0.21	-0.74
4	0.50	-3.16	0.78	-0.74	-5.06	-0.92	0.75	-2.56	-0.98	-0.14	-2.20	-1.14
Panel B: remaining bias after feasible correction (×100)												
-3				-0.001	-0.001	-0.000	0.000	0.000	0.000	0.000	0.000	0.000
-2				-0.035	-0.019	-0.000	-0.052	-0.028	-0.004	-0.073	-0.037	-0.008
-1				-0.018	-0.003	-0.004	-0.031	-0.009	0.006	-0.045	-0.015	0.008
0				-0.008	-0.007	-0.002	-0.003	-0.005	0.001	-0.002	-0.005	0.001
1				-0.005	-0.006	-0.002	0.000	-0.001	-0.001	0.000	-0.000	-0.002
2				-0.002	-0.004	-0.003	0.002	0.000	-0.002	0.003	0.002	-0.003
3				-0.002	-0.003	-0.012	0.008	0.004	-0.011	0.006	0.008	-0.018
4				-0.044	-0.166	-0.101	0.036	-0.136	-0.095	-0.004	-0.144	-0.095

Notes: The coarse and reference models use $n = 6$ and $n = 120$ bins, with $\theta_{RS} = 3$, $\theta_{EC} = 1.5$, and $\beta = 0.95$. Only parameter biases are multiplied by 100. Eigenvalues of Ψ_P are computed on a 40-bin grid, and $\|\pi_j\|/\|r_j\|$ uses the Fisher norm at θ_0. Panel B reports $\theta_j^e - \theta_0 - \tilde{B}_j^e$, using coarse-model derivatives and a reference solution at θ_j^e.

large relative to sampling uncertainty. We therefore conduct a Monte Carlo experiment with T market observations drawn uniformly from the 400 state configurations, observing the actions of both firms. We estimate the two-step, NPL, and full-solution ML models using the $n = 6$ approximation and apply the corresponding plug-in corrections. Each design uses 300 replications. The first-stage CCPs for the two-step estimator are treated as known, allowing us to isolate numerical approximation effects from first-stage estimation noise.

Table 6 reports mean biases and coverage rates. Under strategic complementarity ($\theta_{RN} = -1$), the approximation bias becomes increasingly consequential as the sample size grows. At $T = 100{,}000$, coverage of nominal 95% intervals for full-solution ML falls to 0.633 for the market-size parameter and 0.460 for the competition parameter, compared with 0.793 and 0.700 for two-step estimation. This ordering mirrors the larger ML biases in the population calculations. The plug-in correction improves the corresponding ML coverage rates to 0.953 and 0.960, respectively.

Under strategic substitutability ($\theta_{RN} = 3$), numerical biases are smaller relative to sampling uncertainty, and coverage for two-step and ML estimation remains close to nominal. The correction consequently has a more limited effect. These results reinforce the central implication of Theorem 3: the inferential importance of approximation error depends on its estimator-specific projection, not solely on the norm of the equilibrium error.

The NPL coverage rates require a separate qualification. The reported intervals use the conventional information matrix rather than an NPL-specific asymptotic variance formula.

As shown by Aguirregabiria and Mira (2007) and further analyzed by Kasahara and Shimotsu (2012), NPL has a distinct sampling distribution that reflects its fixed-point estimation structure. Accordingly, NPL coverage in this exercise combines numerical bias with variance-formula error and should not be interpreted as a direct comparison of numerical robustness. In particular, the low coverage for the entry-cost parameter under strategic substitutability persists after bias correction, indicating that a location adjustment alone does not resolve this inferential distortion. Our comparison of approximation sensitivity across estimators therefore focuses primarily on their biases.

The experiment also identifies a limitation of local equilibrium analysis. For $\theta_{RN} \geq 3$, the spectral radius of Ψ_P exceeds one. The equilibrium CCP mapping is therefore not locally contractive under simple fixed-parameter best-response iteration. This does not, by itself, imply nonconvergence of the NPL estimation algorithm, whose local stability depends on the joint updating of parameters and CCPs (Kasahara and Shimotsu, 2012). The NPL fixed-point and first-order conditions can nevertheless define a locally regular pseudo-true solution when $I - \Psi_P$ is nonsingular, as required by Theorem 3.

We also examined parameterizations with stronger positive equilibrium feedback. Within the region where the coarse and reference solutions could be tracked along the same regular equilibrium branch, the largest positive eigenvalue of Ψ_P was approximately 0.5. More pronounced complementarity and higher entry costs sometimes led the two numerical schemes to select different equilibria. Such cases are outside the scope of the local perturbation analysis, which presumes that approximation changes an equilibrium along a common regular branch rather than altering equilibrium selection. We therefore interpret the experiments as evidence on the local propagation of numerical errors and leave approximation-induced equilibrium selection changes for future work.

Overall, the game results illustrate the additional information supplied by the fixed-point bias representation. The norm of equilibrium error alone does not determine the magnitude or ranking of parameter bias. Strategic interaction transforms the primitive residual, and estimator-specific projections determine which components affect the estimated parameters. The feasible bias formulas capture these differences and provide a local numerical diagnostic that can be applied without repeatedly solving the reference model throughout estimation.

7.3 Application: Rust's bus engine data

The preceding experiments isolate the mechanisms of numerical approximation but do not establish whether discretization has materially affected empirical estimates. We therefore apply the diagnostic of Section 6 to the data and specification of Rust (1987). The Madison Metro data record monthly mileage and engine replacements. We examine two samples emphasized by Rust: group 4 (37 buses, 4,329 bus-months and 33 replacements) and groups 1–4

Table 6: Dynamic Entry Game: Monte Carlo Bias and 95% Coverage

			Mean bias (×100)			Coverage: coarse			Coverage: corrected		
θ_{RN}	T	Estimator	RS	RN	EC	RS	RN	EC	RS	RN	EC
-1	10^4	Two-step	5.91	3.33	-0.59	0.917	0.907	0.910	0.943	0.947	0.907
		NPL	5.94	3.27	-0.44	0.963	0.977	0.913	0.977	0.993	0.917
		ML	6.47	3.59	-0.20	0.917	0.903	0.923	0.950	0.950	0.920
-1	10^5	Two-step	4.57	2.95	-0.03	0.793	0.700	0.933	0.953	0.960	0.930
		NPL	4.77	2.97	0.09	0.833	0.773	0.927	0.987	0.993	0.927
		ML	5.38	3.32	0.27	0.633	0.460	0.937	0.953	0.960	0.930
3	10^4	Two-step	0.27	-0.78	-0.24	0.943	0.947	0.943	0.943	0.953	0.943
		NPL	0.69	-0.20	-0.45	0.910	0.907	0.777	0.903	0.907	0.767
		ML	0.52	0.37	-1.11	0.970	0.950	0.943	0.967	0.950	0.947
3	10^5	Two-step	0.38	-0.23	-0.14	0.967	0.963	0.973	0.953	0.970	0.970
		NPL	0.27	0.02	-0.33	0.887	0.933	0.760	0.883	0.933	0.767
		ML	0.05	0.37	-0.76	0.947	0.977	0.957	0.953	0.983	0.970

Notes: Based on 300 replications with $n = 6$, $\beta = 0.95$, $\theta_{RS} = 3$, and $\theta_{EC} = 1.5$. Each market observation includes both firms' actions. Mean biases are multiplied by 100. Two-step estimation uses known first-stage CCPs; corrections are evaluated at each replication's estimate. Coverage uses conventional information-matrix standard errors, which are generally inappropriate for NPL (Kasahara and Shimotsu, 2012) and may be affected by approximation-induced information inequality. The Monte Carlo standard error of coverage near 0.95 is approximately 0.013.

pooled (104 buses, 8,260 bus-months and 60 replacements).

Rust discretizes mileage since the last replacement into 90 states of 5,000 miles, evaluates maintenance cost $c(x) = 0.001\,\theta_{11}x$ at the bin index, and models monthly mileage changes through a multinomial distribution over $\{0, 1, 2\}$ bin increments, estimated in a first step. We retain his discount factor $\beta = 0.9999$ and estimate the replacement cost RC and maintenance-cost coefficient θ_{11}. Our estimates closely reproduce his Table IX: $(RC, \theta_{11}) = (10.19, 2.37)$ versus $(10.08, 2.29)$ for group 4, and $(9.81, 2.67)$ versus $(9.76, 2.63)$ for the pooled sample. We treat the 90-state specification as the coarse model and also consider Rust's 175-state discretization. The reference model evaluates costs at observed continuous mileage and uses the empirical distribution of mileage increments, solved on a 900-bin (500-mile) grid with off-grid evaluation as in Section 4. The diagnostic therefore captures the combined effects of mileage discretization and transition approximation.

Table 7 reports the coarse model estimates and standard errors, the plug-in bias estimate $\hat{B}_j$ from a single reference solution at $\hat{\theta}_j$, the bias to standard error ratio, and the corrected estimates. Reference model maximum likelihood estimates, computed separately for validation, appear in the final columns.

Three findings stand out. First, the estimated numerical bias is small relative to sampling uncertainty in Rust's original samples. With 90 states, the upward bias in θ_{11} is estimated at 0.075 for group 4 and 0.047 for the pooled sample, corresponding to approximately 3.2% and 1.8% of the respective estimates but only 0.14 and 0.10 conventional standard errors. The replacement-cost bias is below 0.05 standard errors in both samples. The direction

Table 7: Rust (1987) Bus Engine Data: Discretization Bias and Correction

		Rust-type MLE		$\hat{B}_j$		$\hat{B}_j/\text{se}$		Corrected (BC)		Reference MLE	
Sample	States	RC	θ_{11}	RC	θ_{11}	RC	θ_{11}	RC	θ_{11}	RC	θ_{11}
Group 4	90	10.189	2.373	0.010	0.075	0.007	0.140	10.179	2.298	10.146	2.285
		(1.351)	(0.538)							(1.356)	(0.546)
Group 4	175	10.224	2.386	0.086	0.103	0.064	0.192	10.137	2.283	10.146	2.285
		(1.356)	(0.537)							(1.356)	(0.546)
Group 4, $\beta = 0.999$	90	10.127	2.423	0.007	0.074	0.006	0.135	10.119	2.349	10.087	2.336
		(1.331)	(0.545)							(1.336)	(0.554)
Groups 1–4	90	9.811	2.671	−0.041	0.047	−0.047	0.105	9.852	2.624	9.833	2.621
		(0.875)	(0.444)							(0.888)	(0.454)
Groups 1–4	175	9.833	2.672	0.027	0.070	0.030	0.159	9.806	2.602	9.833	2.621
		(0.876)	(0.441)							(0.888)	(0.454)

Notes: Data from Rust (1987); $\beta = 0.9999$ unless indicated otherwise. The coarse MLE uses 90 or 175 mileage bins, with costs evaluated at bin indices and bin-increment probabilities estimated in a first step. Parentheses report information-matrix standard errors, excluding first-step uncertainty. The plug-in correction $\hat{\theta}_j - \hat{B}_j$ uses one reference solution at $\hat{\theta}_j$. The reference MLE uses continuous mileage and empirical increments on a 900-bin grid and is reported for validation. Rust's Table IX estimates are $(10.08, 2.29)$ for group 4 and $(9.76, 2.63)$ for groups 1–4.

of the maintenance-cost bias is consistent with evaluating costs at the lower edge of each mileage bin, which understates costs at a given mileage and tends to raise the estimated cost coefficient.

Second, the same numerical displacement becomes consequential as sampling precision increases. Holding the bias and per-observation precision fixed, the group 4 bias would equal one standard error at approximately 220,000 bus-months, about 50 times the original sample. The corresponding threshold for the pooled specification is approximately 750,000 bus-months. Thus, a grid adequate for the original application need not remain adequate in substantially larger fleet panels.

Third, the plug-in correction captures most of the displacement without repeated reference-model optimization. For the baseline 90-state estimates of θ_{11}, it removes approximately 85% and 94% of the respective gaps to the reference MLE. Simply increasing the number of states does not necessarily improve accuracy: the 175-state specification also changes the estimated bin-increment transition, and its estimated bias in θ_{11} is 0.16–0.19 standard errors. These findings reinforce the nonmonotonic grid-refinement pattern in Table 1 and the value of assessing numerical error in parameter units rather than by grid size alone.

7.4 A dynamic entry game: Aguirregabiria and Mira (2007)

The entry game experiment in Section 7.2 isolates equilibrium feedback. We now examine whether a coarse market-size grid generates substantial bias at sample sizes relevant to

Table 8: Entry Game Calibrated to Aguirregabiria and Mira (2007): Market-Size Discretization Bias

			Pseudo-true bias				First-order prediction				$N^\star$ (firm-obs.)	
θ_{RN}	n	Estimator	RS	RN	EC	FC	RS	RN	EC	FC	RS	FC
1	5	Two-step	−0.147	−0.018	−0.008	−0.152	−0.138	−0.023	0.006	−0.130	1,579	1,202
		NPL	−0.166	−0.001	−0.008	−0.173	−0.157	−0.015	0.007	−0.148	1,241	934
		ML	−0.166	−0.002	−0.008	−0.173	−0.158	−0.015	0.007	−0.149	1,373	1,126
1	10	Two-step	−0.017	−0.007	−0.002	−0.017	−0.014	−0.006	0.002	−0.011	123,728	99,927
		NPL	−0.019	−0.005	−0.002	−0.020	−0.016	−0.006	0.002	−0.013	99,043	76,265
		ML	−0.019	−0.004	−0.002	−0.020	−0.016	−0.005	0.002	−0.013	109,634	90,481
1	20	Two-step	−0.008	0.002	−0.000	−0.009	−0.007	0.002	0.000	−0.008	557,209	376,890
		NPL	−0.009	0.003	−0.000	−0.010	−0.009	0.002	0.000	−0.009	426,894	291,148
		ML	−0.009	0.002	−0.001	−0.010	−0.009	0.002	0.000	−0.008	478,310	359,422
2	5	Two-step	−0.135	−0.041	−0.005	−0.138	−0.127	−0.053	0.004	−0.116	1,588	1,550
		NPL	−0.165	0.002	−0.005	−0.172	−0.157	−0.017	0.006	−0.148	1,068	992
		ML	−0.165	0.000	−0.005	−0.172	−0.158	−0.017	0.006	−0.148	1,305	1,376
2	10	Two-step	−0.015	−0.009	−0.001	−0.015	−0.013	−0.009	0.001	−0.010	132,945	134,421
		NPL	−0.019	−0.004	−0.001	−0.019	−0.016	−0.006	0.002	−0.013	91,610	84,199
		ML	−0.019	−0.003	−0.001	−0.020	−0.016	−0.005	0.002	−0.013	111,007	112,352
2	20	Two-step	−0.008	0.001	−0.000	−0.008	−0.007	0.001	0.000	−0.007	556,440	476,292
		NPL	−0.009	0.003	−0.000	−0.010	−0.009	0.002	0.000	−0.009	358,441	303,966
		ML	−0.009	0.002	−0.000	−0.010	−0.009	0.002	0.000	−0.008	455,716	442,509
3	5	Two-step	−0.130	−0.076	−0.004	−0.129	−0.122	−0.098	0.003	−0.107	1,483	1,849
		NPL	−0.165	0.007	−0.004	−0.173	−0.157	−0.014	0.006	−0.148	924	1,025
		ML	−0.165	0.003	−0.003	−0.173	−0.157	−0.016	0.006	−0.149	1,224	1,571
3	10	Two-step	−0.015	−0.011	−0.001	−0.014	−0.012	−0.013	0.001	−0.009	131,472	162,476
		NPL	−0.018	−0.002	−0.001	−0.019	−0.016	−0.005	0.002	−0.013	83,203	88,656
		ML	−0.019	−0.001	−0.001	−0.020	−0.016	−0.003	0.001	−0.013	107,966	128,143
3	20	Two-step	−0.007	−0.001	−0.000	−0.008	−0.007	−0.002	0.000	−0.006	518,401	564,199
		NPL	−0.010	0.003	−0.000	−0.010	−0.009	0.002	0.000	−0.009	306,108	312,893
		ML	−0.009	0.001	−0.000	−0.010	−0.009	0.001	0.000	−0.008	429,453	511,301

Notes: Two symmetric firms; $(\theta_{RS}, \theta_{EC}, \theta_{FC}) = (1, 1, 1.7)$, $\beta = 0.95$, and $\ln S' = 0.2 + 0.8 \ln S + 0.3\varepsilon'$, where $\varepsilon' \sim N(0, 1)$. Coarse models use n-node Tauchen discretization and nearest-node assignment; the reference model uses 200 nodes and local-equilibrium evaluation. Entries report population pseudo-true biases $\theta_j^e - \theta_0$ and first-order predictions from Theorem 3 evaluated at the calibration parameters. The threshold $N^\star = (\mathrm{se}_1/|\theta_j^e - \theta_0|)^2$ is the number of firm observations at which the bias equals one benchmark standard error. The per-observation standard error se_1 uses the relevant exact-model information matrix for ML and two-step estimation with known first-step CCPs; for NPL, the two-step value is used as an approximation. One market-period contributes two firm observations; NPL thresholds are therefore indicative.

empirical entry studies. We adopt the payoff structure of Aguirregabiria and Mira (2007): an active firm receives $\theta_{RS} \ln S_t - \theta_{RN} \ln(1 + a_{-i,t}) - \theta_{EC}(1 - k_{it}) - \theta_{FC}$, plus a type I extreme-value shock, where S_t denotes market size, $a_{-i,t}$ indicates whether the rival is active, and k_{it} indicates whether firm i was active in the previous period. We consider two symmetric firms, set $(\theta_{RS}, \theta_{EC}, \theta_{FC}) = (1, 1, 1.7)$ and $\beta = 0.95$, and vary $\theta_{RN} \in \{1, 2, 3\}$. All four parameters are estimated.

To capture the discretization faced by researchers observing continuous market size, we let $\ln S_t$ follow an AR(1) process with persistence 0.8, unconditional mean 1, and innovation standard deviation 0.3. The coarse transition is constructed by Tauchen's method with $n \in \{5, 10, 20\}$ nodes spanning three unconditional standard deviations; observed market sizes are

assigned to their nearest nodes. The reference model uses 200 nodes with local equilibrium evaluation at observed market sizes. Population criteria are evaluated over 100 quantiles of the stationary market-size distribution crossed with four incumbency configurations, allowing the numerical effects to be isolated from sampling variation.

Table 8 compares the population biases of two-step, NPL, and full-solution ML with the first order predictions from Theorem 3. It also reports $N^\star$, the number of firm observations at which a given bias equals one benchmark standard error. For ML and two-step estimation with a known first step, the benchmarks use their corresponding exact-model information matrices; for NPL, the two-step standard error serves as a proxy, so its thresholds are indicative rather than formal NPL inference calculations.

With five nodes, the market-size coefficient is underestimated by 13–17% and the fixed cost coefficient by about 8–10% across the three estimators. The corresponding $N^\star$ values range from approximately 900 to 1,900 firm observations, or approximately 450 to 950 market periods, with two firms observed in each period. Consequently, these discretization biases can exceed conventional sampling uncertainty even in moderately sized datasets. With ten nodes, the market-size bias falls to 1.5–1.9% and the fixed cost bias to about 0.8–1.2%; the benchmark thresholds increase to roughly 76,000–162,000 firm observations; with twenty nodes they range from approximately 291,000 to 564,000.

The similar directions and magnitudes of the five-node biases suggest that much of the distortion arises from coarse assignment of the continuous market size state. Equilibrium feedback nevertheless changes its magnitude and distribution across parameters: NPL and ML generally have larger market size and fixed cost biases than two-step estimation, whereas two-step exhibits larger biases in the competition parameter. At five nodes, the first-order formulas capture roughly 80–95% of the market size and fixed cost biases; predictions for parameters with much smaller biases are less stable. The evidence therefore favors reporting numerical-bias diagnostics relative to standard errors rather than treating a particular grid resolution as universally adequate. In this calibration, ten nodes substantially reduce the distortion at sample sizes of a few thousand market-periods.

8 Conclusion

Numerical accuracy of a structural model does not directly determine the accuracy of the parameters estimated from it. This paper characterizes the relationship by tracing primitive numerical errors through structural fixed points to their effects on parameter estimation. For likelihood-based estimation, normalization yields a parameter-displacement bound of the same order as the log-likelihood approximation error, without requiring derivative accuracy. Building on the score-projection principle, we identify how Bellman and equilibrium

residuals generate the components of likelihood distortion that matter for parameter bias. In single-agent dynamic discrete choice models, the relevant relative-value errors can remain bounded as the discount factor approaches one under suitable mixing conditions. In dynamic games, strategic feedback transforms numerical errors, while two-step, NPL, and full-solution estimators respond to different projections of those errors. Consequently, even with the same approximate equilibrium mapping, there is no general ranking of estimators by their sensitivity to numerical approximation.

These results have a direct implication for computation and inference. Numerical precision should be assessed in terms of its effects on individual parameter estimates relative to their sampling uncertainty, rather than solely through solver tolerances, likelihood discrepancies, or global value-function errors. Large approximation errors in the structural solution may have little effect on estimation, while smaller errors can lead to consequential bias when aligned with parameter-sensitive directions. Our fixed-point representations provide a basis for diagnosing these effects and allocating computational effort accordingly. Building on existing numerical bias-correction methods (Kristensen and Salanié, 2017), we also show how accurate model evaluations near a coarse estimate can substantially reduce approximation bias without requiring high-accuracy solutions throughout optimization. The numerical experiments illustrate the resulting differences in estimator sensitivity and the implications for confidence interval coverage. The two applications demonstrate the diagnostic's practical value. It confirms that Rust's discretization was adequate for his sample while identifying the sample sizes at which numerical bias becomes consequential. In contrast, it reveals that coarse market-size grids in dynamic entry models can generate substantial parameter bias even at sample sizes typical of empirical applications.

Several extensions remain important. Simulation-based and nonsmooth approximations require further analysis of numerical uncertainty and its interaction with sampling variation (Pakes and Pollard, 1989; Newey and McFadden, 1994; Kristensen et al., 2021). Similar parameter-specific diagnostics could be developed for numerical errors in demand inversion and other moment-based estimators (Dubé et al., 2012). Our dynamic-game results also concern local perturbations around a regular equilibrium; approximation errors near equilibrium multiplicity or instability may affect equilibrium selection itself. These extensions share the central question of this paper: not simply how accurately a structural model is solved, but how accurately its parameters can be estimated and inference conducted using that solution.

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Appendix

A Mathematical Proofs

Throughout, C denotes a generic positive constant independent of j and T, but possibly depending on β unless uniformity in β is explicitly stated. Write $p_\theta(y \mid x) = p(y \mid x; \theta)$ and $p_{j,\theta}(y \mid x) = p_j(y \mid x; \theta)$.

Lemma 1. *Let $g_{j,\theta} = \log p_{j,\theta} - \log p_\theta$ and $D_j(\theta) = Q_j(\theta) - Q(\theta)$. Under Assumption 1(v), for every x and θ, $-\frac{e}{2}\delta_j^2 \le \int p_\theta g_{j,\theta}\, d\mu \le 0$. If Assumption 1(iv) also holds, then, for $\|\theta - \theta_0\| \le \varepsilon_0$,*

$$-L\delta_j\|\theta - \theta_0\| - \frac{e}{2}\delta_j^2 \le D_j(\theta) \le L\delta_j\|\theta - \theta_0\|,$$

and $-\frac{e}{2}\delta_j^2 \le D_j(\theta_0) \le 0$.

Proof. Since $p_{j,\theta} = p_\theta e^{g_{j,\theta}}$ and both densities integrate to one, $\int p_\theta g_{j,\theta}\, d\mu = -\int p_\theta (e^{g_{j,\theta}} - 1 - g_{j,\theta})\, d\mu$. For $|g_{j,\theta}| \le \delta_j \le 1$, $0 \le e^{g_{j,\theta}} - 1 - g_{j,\theta} \le \frac{e}{2} g_{j,\theta}^2 \le \frac{e}{2}\delta_j^2$, which proves the first claim. Next, $D_j(\theta) = \mathbb{E}_X\left[\int p_{\theta_0} g_{j,\theta}\, d\mu\right] = \mathbb{E}_X\left[\int (p_{\theta_0} - p_\theta) g_{j,\theta}\, d\mu\right] + \mathbb{E}_X\left[\int p_\theta g_{j,\theta}\, d\mu\right]$, where the densities are evaluated at X. By Assumption 1(iv), the first term is bounded in absolute value by $L\delta_j\|\theta - \theta_0\|$, while the second lies in $[-e\delta_j^2/2, 0]$. At $\theta = \theta_0$, the first term vanishes. □

Proof of Theorem 1

Proof. Since $|D_j(\theta)| \le \delta_j$ uniformly and $Q_j(\theta_j) \ge Q_j(\theta_0)$, we have $Q(\theta_0) - Q(\theta_j) \le 2\delta_j < \eta(\varepsilon_0)$. Thus $\Delta = \|\theta_j - \theta_0\| < \varepsilon_0$ by Assumption 1(ii). Using local quadratic identification, optimality, and Lemma 1,

$$\lambda\Delta^2 \le Q(\theta_0) - Q(\theta_j) \le D_j(\theta_j) - D_j(\theta_0) \le L\delta_j\Delta + \frac{e}{2}\delta_j^2.$$

Solving this quadratic inequality gives $\Delta \le [L + \sqrt{L^2 + 2e\lambda}]\delta_j/(2\lambda)$, as claimed. □

Proof of Proposition 1

Proof. Optimality gives $\bar{Q}(\theta_0) - \bar{Q}(\bar{\theta}_j) \le 2\delta < \eta(\varepsilon_0)$, so $\|\bar{\theta}_j - \theta_0\| < \varepsilon_0$. Local quadratic identification then yields $\lambda\|\bar{\theta}_j - \theta_0\|^2 \le 2\delta$, proving the bound.

To establish sharpness, consider the example (1). Since $0 \le \phi \le 1$ and $\phi(0) = 1$, we have $\sup_\theta |\bar{Q}_j(\theta) - \bar{Q}(\theta)| = \delta$. Outside $[\sqrt{\delta}/2, 3\sqrt{\delta}/2]$, $\bar{Q}_j(\theta) = -\theta^2/2 \le 0$, whereas $\bar{Q}_j(\sqrt{\delta}) = \delta/2 > 0$. Within this interval, $\bar{Q}_j$ is strictly increasing up to $\sqrt{\delta}$ and strictly decreasing thereafter. Thus the unique maximizer is $\bar{\theta}_j = \sqrt{\delta}$, establishing sharpness. □

Proof of Theorem 2

Proof. By Theorem 1, $\Delta_j = \theta_j - \theta_0 = O(\delta_j)$, so θ_j is interior for sufficiently large j and $\nabla Q_j(\theta_j) = 0$. Let $\bar{H}_j = \int_0^1 \nabla^2 Q_j(\theta_0 + t\Delta_j)\, dt$. A Taylor expansion gives $0 = \nabla Q_j(\theta_0) + \bar{H}_j \Delta_j$. By Assumption 2, $\bar{H}_j = -I + O(\delta_j + \delta_j'')$, and hence $-\bar{H}_j^{-1} = I^{-1} + O(\delta_j + \delta_j'')$.

Using the normalization identity $\int p_{j,\theta_0} \nabla_\theta \log p_{j,\theta_0}\, d\mu = 0$, we obtain

$$\nabla Q_j(\theta_0) = -\mathbb{E}_X \left[\int p_{\theta_0}(e^{g_{j,\theta_0}} - 1) \nabla_\theta \log p_{j,\theta_0}\, d\mu \right] = -\mathbb{E}[s(W) g_j(W)] + O(\delta_j^2 + \delta_j \delta_j').$$

The second equality follows from $e^{g_{j,\theta_0}} - 1 = g_{j,\theta_0} + O(\delta_j^2)$, the bound on the score approximation error, and $\mathbb{E}[\|s(W)\|] < \infty$. Since $\|\mathbb{E}[s(W) g_j(W)]\| = O(\delta_j)$, it follows that

$$\theta_j - \theta_0 = -I^{-1} \mathbb{E}[s(W) g_j(W)] + O\big(\delta_j(\delta_j + \delta_j' + \delta_j'')\big).$$

Also, $\pi_j / p_{\theta_0} = e^{g_{j,\theta_0}} - 1 = g_{j,\theta_0} + O(\delta_j^2)$, which gives the alternative representation of B_j. □

Proof of Corollary 1

Proof. By normalization of p_{j,θ_0}, the identity $\nabla Q_j(\theta_0) = -\mathbb{E}_X[\int \pi_j(y \mid X) s_j(y, X)\, d\mu(y)]$ holds exactly. Since $|\pi_j| \le (e^{\delta_j} - 1) p_{\theta_0}$ and $\mathbb{E}[\|s_j(W)\|] = O(1)$, we have $\|\nabla Q_j(\theta_0)\| = O(\delta_j)$. By Theorem 1, $\Delta_j = \theta_j - \theta_0 = O(\delta_j)$. The uniform Lipschitz condition on $\nabla^2 Q_j$ gives

$$0 = \nabla Q_j(\theta_j) = \nabla Q_j(\theta_0) - I_j \Delta_j + O(\delta_j^2).$$

Since $I_j \to I \succ 0$, I_j^{-1} is uniformly bounded for sufficiently large j. Therefore, $\Delta_j = I_j^{-1} \nabla Q_j(\theta_0) + O(\delta_j^2) = \tilde{B}_j + O(\delta_j^2)$, as claimed. □

Proof of Proposition 2

Proof. (a) The function $W_{\theta_0}(V)(x')$ is continuously differentiable in $V(x', \cdot)$, with gradient $\beta P(\cdot \mid x'; V)$. The integral mean value theorem therefore gives

$$W_{\theta_0}(V_j)(x') - W_{\theta_0}(V)(x') = \beta \sum_{a'} \bar{P}_j(a' \mid x') e_j(x', a').$$

Since $\bar{P}_j$ is a probability vector, $\bar{M}_j$ is a Markov kernel. The fixed-point equations imply $e_j = \beta \bar{M}_j e_j + r_j$, and the Neumann series converges because $\beta < 1$. For the linearization, let $q_j = (\bar{M}_j - M_j) e_j$. Since $\sum_{a'} (\bar{P}_j - P_{\theta_0})(a' \mid x') = 0$,

$$q_j(x, a) = \int \sum_{a'} (\bar{P}_j - P_{\theta_0})(a' \mid x') \{e_j(x', a') - e_j(x', 0)\} K_j(dx' \mid x, a).$$

The logit Lipschitz property gives $\sup_{x'} \|\bar{P}_j - P_{\theta_0}\|_1 \le C \operatorname{osc}(e_j)$, so $\|q_j\|_\infty \le C \operatorname{osc}(e_j)^2$. Hence $R_j = \beta(I - \beta M_j)^{-1} q_j$. Using $\operatorname{osc}(h) \le \operatorname{sp}(h)$ and the span-contraction inequality yields $\operatorname{osc}(R_j) \le C\beta \operatorname{osc}(e_j)^2 \sum_{k\ge 0} \beta^k \tau(M_j^k)$. The sum is finite for fixed $\beta < 1$, and the bound is uniform in β under the stated mixing condition.

(b) Since $v_j - v = \beta e_j$, the logit log-likelihood distortion is

$$g_j(a, x; \theta_0) = \beta e_j(x, a) - \{\mathrm{LSE}(v_j(x, \cdot)) - \mathrm{LSE}(v(x, \cdot))\},$$

where $\mathrm{LSE}(v) = \log \sum_b e^{v_b}$. By monotonicity and translation equivariance of LSE, the term in braces lies between $\beta \min_b e_j(x, b)$ and $\beta \max_b e_j(x, b)$. Thus $|g_j(a, x; \theta_0)| \le \beta \operatorname{osc}(e_j)$.

(c) Since sp is a seminorm and $\operatorname{sp}(\bar{M}_j^k r_j) \le \tau(\bar{M}_j^k) \operatorname{sp}(r_j)$, the Neumann series gives $\operatorname{sp}(e_j) \le \operatorname{sp}(r_j) \sum_{k\ge 0} \beta^k \tau(\bar{M}_j^k)$. Under the m-step Doeblin condition, $\tau(\bar{M}_j^m) \le 1 - \rho$. Submultiplicativity implies $\tau(\bar{M}_j^k) \le (1 - \rho)^{\lfloor k/m \rfloor}$, and therefore

$$\sum_{k\ge 0} \beta^k \tau(\bar{M}_j^k) \le \frac{\sum_{i=0}^{m-1} \beta^i}{1 - \beta^m(1 - \rho)} \le \frac{m}{1 - \beta^m(1 - \rho)} \le \frac{m}{\rho}.$$

This proves the stated bounds.

(d) Since K_j and f are probability measures, $\int c\,(K_j - f) = 0$. Thus $r_j = \int \{W_{\theta_0}(V) - c\}(K_j - f)$, and the result follows from $\operatorname{sp}(r_j) \le 2 \sup_{x,a} |r_j(x, a)|$. □

Proof of Theorem 3

Proof. Write $d_j = \delta_{\Psi,j}$, $G = (I - \Psi_P)^{-1}$, and $D = \partial_\theta P = G\Psi_\theta$. The nonsingularity assumptions and the implicit function theorem applied to the respective population first-order and equilibrium conditions imply $\Delta_j^e = \theta_j^e - \theta_0 = O(d_j)$ for $e \in \{2S, NPL, ML\}$. The equilibrium CCP displacements associated with NPL and full-solution ML are also $O(d_j)$. For any normalized conditional probability vector $\Pi(\theta)$, the population score satisfies

$$\mathbb{E}[\nabla_\theta \log \Pi(\theta)(Y \mid X)] = \mathbb{E}_X \left[\sum_{a \in A} \{P_0 - \Pi(\theta)\}(a \mid X) \frac{\nabla_\theta \Pi(\theta)(a \mid X)}{\Pi(\theta)(a \mid X)} \right]. \tag{5}$$

For a multi-player CCP profile, the identity follows by applying it to each player's normalized CCPs and summing over players, equivalently using the induced joint likelihood. All expansions below have $O(d_j^2)$ remainders by the assumed C^2 accuracy and smoothness. Since the state space is finite and P_0 is a strictly positive equilibrium profile, P_0 is bounded away from zero, and each $\Pi(\theta_j^e)$ below satisfies $\Pi(\theta_j^e) = P_0 + O(d_j)$ and therefore inherits a uniform positive lower bound for j large. Consequently the weights $1/\Pi(\theta_j^e)$ appearing in (5) differ from the equilibrium weights $1/P_0$ defining $\langle \cdot, \cdot \rangle$ by $O(d_j)$, and since the discrepancy

$P_0 - \Pi(\theta_j^e)$ is itself $O(d_j)$, replacing the former by the latter contributes $O(d_j^2)$. We therefore write all first-order conditions directly in terms of $\langle \cdot, \cdot \rangle$.

First, expanding the approximate equilibrium condition $P_j(\theta_0) = \Psi_j(P_j(\theta_0), \theta_0)$ around P_0 gives $\pi_j = \Psi_P \pi_j + r_j + O(d_j^2)$, where the remainder collects the second-order Taylor term in π_j and the cross term $(\Psi_{j,P} - \Psi_P)\pi_j$, both of which are $O(d_j) \cdot O(d_j) = O(d_j^2)$ by C^1 accuracy and $\pi_j = O(d_j)$. Hence $\pi_j = G r_j + O(d_j^2)$. In each expansion below, the remainder consists of second-order Taylor terms and products of $O(d_j)$ derivative discrepancies with $O(d_j)$ displacements, and is therefore $O(d_j^2)$; we do not repeat this argument.

Two-step: With the first-step CCPs fixed at P_0, $\Psi_j(P_0, \theta_j^{2S}) - P_0 = r_j + \Psi_\theta \Delta_j^{2S} + O(d_j^2)$. Moreover, $\Psi_{j,\theta}(P_0, \theta_j^{2S}) = \Psi_\theta + O(d_j)$. Applying (5) yields $0 = -\langle \Psi_\theta, r_j + \Psi_\theta \Delta_j^{2S} \rangle + O(d_j^2)$, which gives B_j^{2S}.

NPL. Let $u_j = P_j^{NPL} - P_0$ and $\Delta_j^{NPL} = \theta_j^{NPL} - \theta_0$, both $O(d_j)$ by the implicit function theorem. Expanding the equilibrium restriction $P_j^{NPL} = \Psi_j(P_j^{NPL}, \theta_j^{NPL})$ around (P_0, θ_0) gives $u_j = \Psi_P u_j + \Psi_\theta \Delta_j^{NPL} + r_j + O(d_j^2)$, and hence $u_j = D\Delta_j^{NPL} + \pi_j + O(d_j^2)$ by $\pi_j = G r_j + O(d_j^2)$ and $D = G\Psi_\theta$. The NPL first-order condition differentiates the pseudo-likelihood with respect to θ holding the conjectured profile fixed at P_j^{NPL}, so it is (5) with $\Pi(\theta) = \Psi_j(P_j^{NPL}, \theta)$ evaluated at $\theta = \theta_j^{NPL}$. There $\Pi(\theta_j^{NPL}) = P_j^{NPL}$, so $P_0 - \Pi(\theta_j^{NPL}) = -u_j$, and $\nabla_\theta \Pi(\theta_j^{NPL}) = \Psi_{j,\theta}(P_j^{NPL}, \theta_j^{NPL}) = \Psi_\theta + O(d_j)$ by C^1 accuracy of Ψ_j and $\|(u_j, \Delta_j^{NPL})\| = O(d_j)$. Substituting into (5) and using $u_j = O(d_j)$, $0 = -\langle \Psi_\theta, u_j \rangle + O(d_j^2) = -\langle \Psi_\theta, D\Delta_j^{NPL} + \pi_j \rangle + O(d_j^2)$, which yields B_j^{NPL}.

Full-solution ML: The approximate equilibrium profile satisfies $P_j(\theta_j^{ML}) - P_0 = \pi_j + D\Delta_j^{ML} + O(d_j^2)$ and $\partial_\theta P_j(\theta_j^{ML}) = D + O(d_j)$. The likelihood first-order condition therefore becomes $0 = -\langle D, \pi_j + D\Delta_j^{ML} \rangle + O(d_j^2)$, giving B_j^{ML}.

Solving the three first-order conditions under the nonsingularity assumptions yields $\Delta_j^e = B_j^e + O(d_j^2)$. Finally, the assumed C^2 approximation accuracy and local regularity imply $\Psi_{j,\theta} - \Psi_\theta = O(d_j)$ and $\partial_\theta P_j - D = O(d_j)$. Since $r_j = O(d_j)$ by definition and $\pi_j = (I - \Psi_P)^{-1} r_j + O(d_j^2) = O(d_j)$, replacing the exact derivatives by their approximate-model counterparts changes the bias formulas only by $O(d_j^2)$, proving the final claim. □

Proof of Theorem 4

Proof. (i) Since $\sup_\theta |\hat{Q}_{j_T}(\theta) - \hat{Q}(\theta)| \leq \delta_T$ deterministically and $\sup_\theta |\hat{Q}(\theta) - Q(\theta)| = o_p(1)$ by Assumption 3(i), we have $\sup_\theta |\hat{Q}_{j_T}(\theta) - Q(\theta)| = o_p(1)$. Consistency follows from the identification condition and the extremum estimator consistency theorem of Newey and McFadden (1994).

(ii) Write $j = j_T$. By (i) and Theorem 1, both $\hat{\theta}_j$ and θ_j lie in the interior neighborhood of θ_0 with probability approaching one. Define $\bar{H}_j = \int_0^1 \nabla_\theta^2 \hat{Q}_j\big(\theta_j + t(\hat{\theta}_j - \theta_j)\big)\, dt$. The sample first-order condition gives $\sqrt{T}(\hat{\theta}_j - \theta_j) = -\bar{H}_j^{-1} \sqrt{T} \nabla_\theta \hat{Q}_j(\theta_j)$. Since the approximate and

exact sample Hessians differ uniformly by at most δ_T'', the uniform law of large numbers and consistency imply $\bar{H}_j \to_p -I$ and $\hat{I}_j \to_p I$.

For the score, let $d_j(W) = \nabla_\theta \log p_j(Y \mid X; \theta_j) - s(W)$. Both scores have mean zero by their respective population first-order conditions. Moreover,

$$\mathbb{E}[\|d_j(W)\|^2] \le 2(\delta_T')^2 + 2\mathbb{E}\left[\sup_{\|\theta-\theta_0\| \le \|\theta_j-\theta_0\|} \|\nabla_\theta \log p(Y \mid X; \theta) - s(W)\|^2\right] \to 0,$$

by $\theta_j \to \theta_0$, continuity of the exact score, and dominated convergence under Assumption 3(ii). Hence $T^{-1/2}\sum_{t=1}^T d_j(W_t) = o_p(1)$, and therefore $\sqrt{T}\nabla_\theta \hat{Q}_j(\theta_j) = T^{-1/2}\sum_{t=1}^T s(W_t) + o_p(1) \Rightarrow N(0, I)$. Together with $\bar{H}_j \to_p -I$, this establishes (ii).

(iii) By Theorem 2, $\theta_j - \theta_0 = B_j + O(\delta_T(\delta_T + \delta_T' + \delta_T''))$. Under the stated rate condition, the remainder is $o(T^{-1/2})$, so (ii) implies $\sqrt{T}(\hat{\theta}_j - \theta_0 - B_j) \Rightarrow N(0, I^{-1})$. If $\sqrt{T}\delta_T \to 0$, Theorem 1 gives $\sqrt{T}(\theta_j - \theta_0) \to 0$, so approximation is asymptotically negligible. If instead $\sqrt{T}B_j \to b$, then $\sqrt{T}(\hat{\theta}_j - \theta_0) \Rightarrow N(b, I^{-1})$. For component k, let $\sigma_k^2 = (I^{-1})_{kk}$. The corresponding Wald statistic converges to $N(b_k/\sigma_k, 1)$, implying coverage strictly below the nominal level for a two-sided symmetric confidence interval whenever $b_k \neq 0$. □

Proof of Theorem 5

Proof. (i) Since $\nabla \hat{Q}_{j^\star}(\hat{\theta}_{j^\star}) = 0$, the integrated Hessian expansion gives $\nabla \hat{Q}_{j^\star}(\hat{\theta}_j) = -\tilde{I}(\hat{\theta}_j - \hat{\theta}_{j^\star})$ where $\tilde{I} = -\int_0^1 \nabla_\theta^2 \hat{Q}_{j^\star}\big(\hat{\theta}_{j^\star} + t(\hat{\theta}_j - \hat{\theta}_{j^\star})\big)\, dt$. Consequently, the Newton correction satisfies exactly

$$\hat{\theta}_j^{NR} - \hat{\theta}_{j^\star} = \hat{I}_j^{-1}(\hat{I}_j - \tilde{I})(\hat{\theta}_j - \hat{\theta}_{j^\star}).$$

The uniform Hessian approximation bounds and local sample-Hessian Lipschitz condition imply $\|\hat{I}_j - \tilde{I}\| = O_p(\delta_j'' + \|\hat{\theta}_j - \hat{\theta}_{j^\star}\|)$. By Theorem 1, $\theta_j - \theta_{j^\star} = O(\delta_j)$. Moreover, the common influence-function expansion established in Theorem 4(ii) gives $\hat{\theta}_j - \hat{\theta}_{j^\star} = O_p(\delta_j) + o_p(T^{-1/2})$. Since $\hat{I}_j^{-1} = O_p(1)$, it follows that $\hat{\theta}_j^{NR} - \hat{\theta}_{j^\star} = O_p(\delta_j \kappa_j) + o_p(T^{-1/2})$.

(ii) Let $\tilde{B}_j$ denote the population correction in Corollary 1. Decompose $\hat{\theta}_j^{BC} - \theta_0 - (\hat{\theta}_j - \theta_j) = (\theta_j - \theta_0 - \tilde{B}_j) + (\tilde{B}_j - \hat{B}_j)$. By Corollary 1, the first term is $O(\delta_j^2)$.

For the second term, write $\tilde{B}_j = -I_j^{-1}F_j(\theta_0)$ and $\hat{B}_j = -\hat{I}_j^{-1}\hat{F}_{T,j}(\hat{\theta}_j)$, where $F_j(\theta) = \mathbb{E}_X[\sum_a \nabla_\theta \log P_j(a \mid X; \theta)\{P_j - P\}(a \mid X; \theta)]$ is the population CCP-projection term and $\hat{F}_{T,j}$ its sample analogue with P replaced by $P_{j^\star}$, as in condition (b). Then $\tilde{B}_j - \hat{B}_j = -(I_j^{-1} - \hat{I}_j^{-1})F_j(\theta_0) - \hat{I}_j^{-1}\{F_j(\theta_0) - \hat{F}_{T,j}(\hat{\theta}_j)\}$. Since $|\pi_j| \le (e^{\delta_j} - 1)p_{\theta_0}$, we have $F_j(\theta_0) = O(\delta_j)$. The sample-Hessian conditions in (a) give $\hat{I}_j - I_j = O_p(\|\hat{\theta}_j - \theta_0\| + T^{-1/2}) = O_p(\delta_j + T^{-1/2})$, so the first term is $O_p(\delta_j(\delta_j + T^{-1/2}))$, which is dominated by the bound below. For the second term, $\hat{I}_j^{-1} = O_p(1)$, and $F_j(\theta_0) - \hat{F}_{T,j}(\hat{\theta}_j)$ consists of three pieces: replacing the exact CCPs by the reference approximation contributes $O(\delta_{j^\star})$; evaluating at $\hat{\theta}_j$ rather than θ_0

contributes $O_p((\delta_j + \delta'_j)\|\hat{\theta}_j - \theta_0\|) = O_p((\delta_j + \delta'_j)(\delta_j + T^{-1/2}))$ by the Lipschitz modulus in (b); and the sampling error of the empirical projection at a fixed θ is $O_p(\delta_j T^{-1/2})$ because its summands are $O(\delta_j)$. Collecting terms, $\tilde{B}_j - \hat{B}_j = O_p(\delta_j \kappa_j + \delta_{j^\star} + (\delta_j + \delta'_j)T^{-1/2})$, which establishes (ii). Finally, under the stated rate conditions, the remainder terms in both corrections are $o_p(T^{-1/2})$. Theorem 4(ii)–(iii) and Slutsky's theorem yield the asserted asymptotic distributions. □

Proof of Theorem 6

Proof. Write $d_j = \delta_{\Psi,j}$, $d_{j^\star} = \delta_{\Psi,j^\star}$, and $q_T = d_j + a_T + T^{-1/2}$. By Theorem 3, $\theta_j^e - \theta_0 = B_j^e + O(d_j^2)$, with $r_j, \pi_j = O(d_j)$. By Assumption 4(ii), $\hat{\theta}_j^e - \theta_0 = O_p(d_j + T^{-1/2})$ for $e \in \{2S, NPL\}$.

Uniform C^1 approximation accuracy, first-step consistency, and local regularity imply $\hat{r}_j - r_j = O_p(d_j q_T + d_{j^\star})$ and $\hat{\pi}_j - \pi_j = O_p(d_j q_T + d_{j^\star})$. For the latter, the implicit function theorem gives $\|P_j - P\|_{C^1} = O(d_j)$ and $\|P_{j^\star} - P\|_{C^1} = O(d_{j^\star})$. Similarly, the estimated derivative matrices A_j^{2S}, A_j^{NPL}, and D_j^{NPL} differ from their exact counterparts by $O_p(q_T)$. Since the state and action spaces are finite and CCPs are bounded away from zero, empirical Fisher inner products have $O_p(T^{-1/2})$ sampling errors. The nonsingularity assumptions ensure that the estimated matrices have bounded inverses with probability approaching one. As $r_j, \pi_j = O(d_j)$, standard matrix perturbation bounds yield $\hat{B}_j^e - B_j^e = O_p(d_j^2 + d_{j^\star} + d_j(a_T + T^{-1/2}))$. Finally, combining this bound with Theorem 3 gives

$$\hat{\theta}_j^{e,BC} - \theta_0 = (\hat{\theta}_j^e - \theta_j^e) + O_p\big(d_j^2 + d_{j^\star} + d_j(a_T + T^{-1/2})\big).$$

Under the stated rate conditions, the remainder is $o_p(T^{-1/2})$. Under the stated rate conditions the remainder is $o_p(T^{-1/2})$, so $\sqrt{T}(\hat{\theta}_j^{e,BC} - \theta_0) = \sqrt{T}(\hat{\theta}_j^e - \theta_j^e) + o_p(1)$, and the result follows from Assumption 4(ii) and Slutsky's theorem. □

B Computational Details

This appendix summarizes the numerical implementation of Section 7. Throughout, approximation errors are measured relative to a substantially finer reference solution rather than an analytically exact solution. For the engine-replacement model, the state space is discretized into equal-width bins, with transition probabilities computed analytically from the exponential distribution, including the boundary atom at $x = 1$. The value function is evaluated at bin midpoints, and off-grid continuation values are obtained by one application of the discretized operator in (3). The fixed point is solved by Newton's method to a residual tolerance of 10^{-13}. The reference model uses $n = 1000$ bins. Population expectations are evaluated over 2,000 equally spaced states, and pseudo-true parameters are obtained

by maximizing the population criterion. Derivatives are computed by central differences. Monte Carlo samples are drawn from the reference model, with 300 replications per design. The plug-in correction uses one reference solution at the coarse estimate, while the Newton correction uses finite-difference reference scores.

For the dynamic entry game, transition probabilities are computed from the normal CDF, accounting for boundary masses. Given a CCP profile, the best-response mapping is evaluated by solving the corresponding continuation-value linear system and applying the logit choice formula. Equilibria are computed by Newton-type methods, using nearby solutions as initial values. The coarse and reference models use $n = 6$ and $n = 120$ bins, respectively, with equilibrium residuals below approximately 10^{-12}. At off-grid states, local equilibrium CCPs are solved using the grid-based continuation values, ensuring a common approximation scheme across estimators. Population calculations use 100 market-size points and four incumbency configurations. The two-step estimator uses true first-stage CCPs, while NPL pseudo-true parameters are obtained by jointly solving the fixed-point restrictions and first-order conditions. Derivatives are computed numerically, and Monte Carlo results use 300 replications per design.

For the application to Rust (1987), we parse the original odometer files following their documented layout, construct mileage since replacement from odometer readings and recorded replacement mileages, and assign each replacement decision to the month preceding the reset. Following Rust's Table IX, we estimate bin-increment probabilities in a first step and maximize the partial likelihood for (RC, θ_{11}) by Fisher scoring, solving the value function by Newton's method to a tolerance of 10^{-12}. The reference model uses the empirical distribution of mileage increments, solves the value function on a 900-bin grid with 500-mile spacing, and evaluates costs and choice probabilities at observed mileage using one application of the operator. For the calibrated entry game, Tauchen's grid spans three unconditional standard deviations of $\ln S$. Observed market sizes are assigned to the nearest grid point in the coarse model and evaluated using local equilibrium in the 200-node reference model. Population pseudo-true values for the design of Section 7.4 are computed as in Section 7.2, and benchmark standard errors are obtained from the exact-model information matrix in the Fisher metric, using the two-step benchmark as a proxy for NPL.

For the span-bound calculations, we construct $\bar{M}_j$ in Proposition 2(a) from K_j and the integrated CCPs $\bar{P}_j$, approximated using a 64-point midpoint rule. Since K_j is supported on n grid points, powers of $\bar{M}_j$ are computed using an initial transition matrix and a $2n \times 2n$ grid matrix T_j. We compute $\tau(T_j^m)$ as half the maximum pairwise row L^1 distance and obtain state-uniform bounds from Remark 1. The series is summed through $m = 80$, with the remaining tail bounded using $\tau(T_j^5)^{\lfloor m/5 \rfloor}$. Numerical solution tolerances are substantially tighter than the approximation errors reported in the tables.